Atlantium
Enlightened Water Technologies

UV Water Treatment
Atlantium Hydro-Optic™ Solutions

User Manual

Guide to the Atlantium Hydro-Optic UV System

Model RS104

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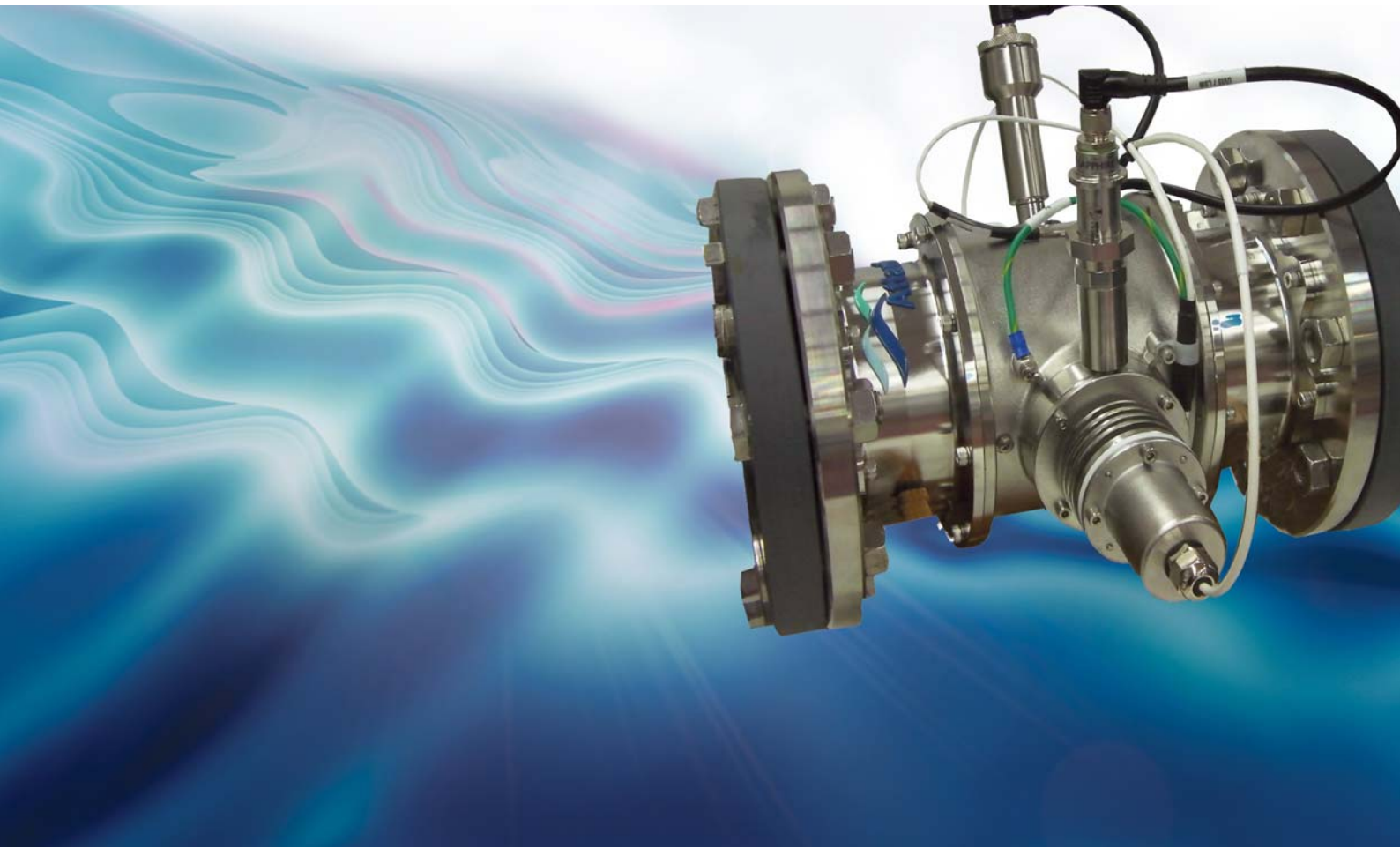

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System and accessory specifications are subject to change without notice.

IMPORTANT NOTICE

It is strictly forbidden to alter/change the system hardware in any way.

Any change made without written permission from Atlantium shall void the warranty.

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003	CIP procedure updates. Added circuit breaker information.	July 2015
004	Wiring correction	Sept 2015
005	Updates and corrections to electrical and warning icons.	Mar 2017
006	PS130000E contains electrical information for both 400VAC & 440/480VAC.	Apr 2017
007	CIP procedure corrections and minor corrections.	Apr 2018
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1 Introduction

Congratulations on purchasing an Atlantium **RS104** Hydro-Optic UV Water Treatment System. This user guide provides you with clear guidelines for correct installation procedures and a good understanding of all system functions, so that you can derive maximum benefit from your Atlantium Unit. The Atlantium **RS104** guarantees water bio-security at optimized energy efficiency, and meets EPA criteria for 4-log virus.

The system is designed to provide applications for the following Industries:

- **Dairy**
- **Bio-Pharma**
- **Municipal**
- **Power Plants**
- **Aquaculture**
- **Bottled Water**
- **Beverages**
- **Beer**
- **Aquariums**
- **and more**

Applications

- **Water Disinfection** - meets highest standards for water disinfection.
- **Chlorine Replacement** - provides a chemical-free alternative to chlorine for treating source water. It easily inactivates microorganisms including those that are resistant to chlorine and heat.
- **Non-Thermal Pasteurization** - replaces energy-intensive heat pasteurization of water - meets US FDA criteria for pasteurized equivalent water,.
- **Post-Active Carbon** - protects against breakthroughs and reduces bio-fouling downstream. Significantly reduces the frequency of aggressive cleaning, and lengthens the lifespan of RO membranes
- **Dechlorination / Chlorine Destruction -** efficient, clean, non-chemical alternative to activated carbon filter or sodium biosulfite. Reduces risk, better protection, lower operating expenses.
- **Ozone Destruction** – efficient, cost-effective solution.
- And more.

Patented Innovative Technology

- Field proven worldwide, used by world-leading brands
- Based on Hydro-Optic engineering that optimizes UV efficiency and enables uniform dose distribution
- Medium Pressure High Intensity UV effective in cold & warm water, compromises microorganisms' repair mechanisms for long-term biosecurity
- Guaranteed required dose for complete microbial inactivation.
- Sustainable operation provides the security and stability that the industry demands.

Guaranteed Dose Delivery

- Unique monitoring & control software assures that required UV dose is being delivered at all times.
- Two sensors per lamp for real-time monitoring
- Automatically adjusts lamp power based on real time analyses of critical parameters such as water flow rate, lamp intensity and water transmittance to UV (UVT).
- A user-friendly interface displays the monitored data in real time including the actual delivered dose as well as the other critical parameters, so you can verify that you are actually getting the dose that you need.

Lower Electrical Consumption

Fiber optic and hydraulic principles applied to UV system design enables:

- Optimized UV power and uniform dose distribution.
- Efficient use of power, with lower electrical consumption than low or medium-pressure-based UV systems.

Minimum Down Time

- True in-line system.
- UV lamps securely positioned outside the water.
- Quick lamp replacement (4 minutes) and better safety.

Integrates with your Process Line

- Field proven installation.
- With no additional software, you can integrate the Atlantium system into your current process using Atlantium's customizable settings.

TRACS Total Reporting Analysis & Control Software (Optional)

- Reporting functions meet your regulatory requirements
- A single installation of Atlantium's **TRAC Software** enables real-time access of up to nine Atlantium systems.
- The sophisticated data logging system enables traceability over time of all monitored data.
- It maintains a database and provides detailed analysis and reporting functions that can be used for trend analysis or proof of compliance.
- Settings reports document any changes to the operations and controls
- You can print out history and all the reports you need, ready for signature.

TRACS Total Reporting Analysis & Control Software can be upgraded into your Atlantium system if the need arises at a later time.

Efficiency and Easy Maintenance

- For total efficiency and dependable performance, the Atlantium system monitors the lamps for performance, assuring that a lamp is replaced only when absolutely needed.
- By following the easy preventive maintenance programs you keep the system working at peak performance.

1.1 Before You Begin

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage to the products
- It is your responsibility to use the product in an appropriate manner.
- This product is designed for use solely in indoor industrial environment or, if expressly permitted, also in the field and must not be used in any way that may cause personal injury or property damage.
- You are responsible if the product is used for any intention other than its designated purpose or in disregard of Atlantium instructions. Atlantium shall assume no responsibility for such use of the product.
- The product is used for its designated purpose if it is used in accordance with its product documentation and within its performance limits.
- Using the product requires technical skills and a basic knowledge of English. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel with the required skills be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the subsequent users.
- Applicable local or national safety regulations and rules for the prevention of accidents must be observed in all work performed.

1.2 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers.

DANGER!

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

WARNING!

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION!

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

ATTENTION!

Indicates the possibility of incorrect operation which can result in damage to the product.

Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.

Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.

Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.

Indicates a hazardous situation involving caustic chemicals which, if not avoided, can result in death or serious damage.

Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.

Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.

Indicates information related to installation, safety or use of system.

2 About the RS104 Systems

This Atlantium **RS104** Hydro Optic Water Disinfection System includes a disinfection unit housing a UV lamp, an electrical cabinet that contains the ballast for the lamp and a **Controller.** Optimized light distribution reflects UV light back into the water to uniformly distribute the UV dose. Water flows unimpeded, resulting in low head-loss.

The specially designed short (Arc length 95mm / 3.74 inch) MPHI lamps deliver a broad spectrum disinfection that irrevocably damages DNA and proteins in microorganisms. Housing Atlantium Quick Connect MPHI lamps within a thick quartz tube significantly reduces risk of breakage and its consequences. A UV Intensity Sensor (UVIS) provides continuous measurement of each lamp's output. In addition, a UVT Analyzer reports and tracks the water quality.

2.1 The RS104 Unit

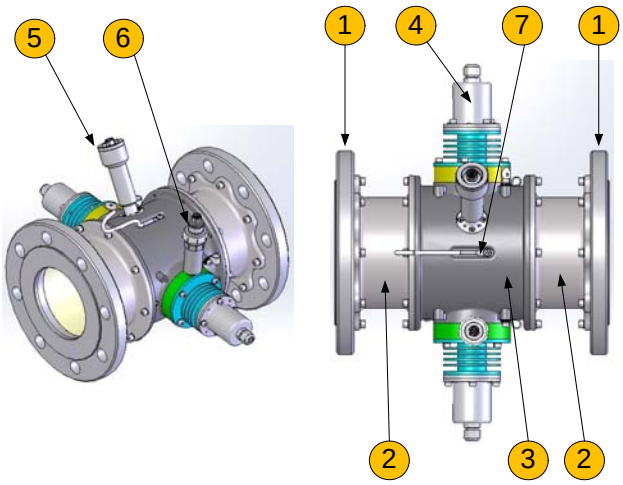


Figure 2-1: The Atlantium **RS104** Unit

Table 2-1: Atlantium Unit Key

#	Item	Description
1	Flange interface	The unit connects to the facility piping via a flange interface as per customer order
2	Disinfect Chamber	The pipe in which the disinfection occurs
3	Lamp Chamber	The chamber that holds the lamp
4	Lamp Arc length (mm / inch): Lamp total length (mm / inch):	Lamp encased in quartz tube assembly Medium Pressure High Intensity (MPHI) 95 / 3.74 262.8 / 10.35

Table 2-1: Atlantium Unit Key

#	Item	Description
5	UVT Sensor	The sensor that measures the UV transmittance through the water. Together with the UV Intensity sensor and the Controller, it analyzes, tracks and reports the water quality.
6	UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds it to the real time dose calculation.
7	Temperature Sensor	The sensor that measures the temperature of the Unit

2.2 RS104 System's Interface Control Drawings

Refer to the accompanying ICD of the **RS104** system.

2.3 Atlantium System Footprint

The Atlantium system has a small, compact footprint that facilitates in-line installation in any process train.

Table 2-2: **RS104** Physical Specifications

Item	Detail		
lamps	1		
Atlantium Unit Construction Materials: Casing: Quartz Lamp Chamber:	 Electro-polished Stainless steel CF8M (316) High grade fused silica (Quartz)		
Standard Pipe Flange Interface Options:	Flange		Ferrule/Tri-Clamp
	DIN2527 DN100 PIN16	ANSI B 16.5 4" 150lb	DIN 32676 DN100
Unit Length (mm / inch)	287.4/11.31	295.4/11.63	323.4/12.73
Inner ∅ Diameter (mm / inch)	100/3.94	101.6/4	100/3.94
Outer ∅ Diameter (mm / inch)	220/8.66	228.6/9	119/4.69
Bolt Circle Diameter	180/7.08	191/7.52	N/A
Bolt Hole Diameter	191/7.52		N/A
Number of Bolts	8		N/A
Width at Widest Point (mm / inch)	405.4 / 16		
Unit Volume (Liter / Gallon)	2.29/0.52		2.56/0.58
Weight (without water) (Kg / lb)	22.4/49.4	25.2/55.6	18.2/40.1
Service Clearance on each side (mm / inch)	450 / 17.72		
Minimum Height above Floor Level (mm / inch)	700 / 27.6		
Maximum Working Pressure	10 Bars / 145 P.S.I.		
Maximum Flow Rate	Application dependent		

2.4 The Electrical Cabinet

The Atlantium system includes a Electrical Cabinet that may be mounted on the wall or on a suitable anchoring point. The location of the Electrical Cabinet must be designated according to the maximum distance needed for positioning the Cable Harness of the Unit.

The length of the standard Cable Harness that connects the Unit to its electrical cabinet is 5 meters (16.5 ft.). This means that you must determine the path that the Cable Harness can be placed and on that basis, calculate where the electrical cabinet can be located. The electrical cabinet can be affixed to a wall or onto a bracket structure.

The electrical cabinet includes air vents with fans and filters on the top and bottom of each side.

Refer to the accompanying ICD of the Electrical Cabinet.

Table 2-3: Electrical Cabinet Physical Details

Physical	Details
Electrical Cabinet (H x W x D) (mm/inch):	641.6 / 25.3 x 425 / 16.7 x 255.4 / 10.1
Weight (Kg / lb):	22.2 / 49
Clearance (mm / inch): Filter Removal: Open Door Radius:	 500 / 19.7 400 / 15.8
Protection Category:	IP 55

2.5 Electrical Usage

Utilize a main circuit breaker for each power feeding type C according to the rating specified in Table 2-4 on page 13.

Where the main source is 440/480 VAC with second source of 100-130 VAC, utilizing two main circuit breakers fits branch circuit protection and meets the UL 489 standard.

Utilize a Residual Current Device (RCD) that is Type A with 3000A surge resistance.

Atlantium recommends ABB P/N F204 A-xx/0.03 AP-R (xx represents the current rating according to the system load. Select the appropriate RCD accordingly. Calculating the value, takes into account the power consumption of the system (see the table in the section below) and the mains voltage of your locale.

The tables below detail the Atlantium Unit's electrical details and power consumption.

Table 2-4: **RS104** Electrical Details - Voltage Requirements

Electrical Parameter	Details
Main power source	400/440/480 VAC 3ph 50/60Hz
Second power source (only for 440/480 main source)	110/120/127 VAC 1ph 50/60Hz
Maximum allowed voltage tolerance:	±10% from the nominal voltage

2.5.1 Power Consumption and Local Voltage Requirements

The table below details the power consumption of the Atlantium system with direct feeding according to local voltage.

# of lamps:		1	
Nominal lamp power (kW):		1	
Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required Circuit Breaker (Type C)
Direct Feeding (w/o Power Module)	400	1.7	3x3A
Direct Feeding (w/o Power Module)	440	1.4	3x3A
	480	1.3	3x3A
	110/120/127	0.5	1x3A

2.6 Environmental Requirements

The Atlantium system is to be situated indoors in an area that is adequately cooled and with sufficient air flow. However, there have been many successful Atlantium installations in less friendly environments. Consult your Atlantium representative to choose the most suitable location.

Table 2-5: Atlantium System Environmental Details

Environmental	Details
Maximum ambient air temperature:	50°C (122°F)
Maximum ambient air temperature in the electrical cabinet's environment: For locations with a higher temperature and relative humidity, Electrical Cabinets with an air conditioning unit are available. Consult with an Atlantium representative.	35°C (95°F)
Maximum atmospheric relative humidity:	80%
Water Temperature: Maximum – high temperature: For water temperature above 70°C (160°F) consult with Atlantium service engineers Minimum – cold temperature:	90°C (194°F) No limit

3 Getting Ready for Installation

Using the guidelines and information in this section, you can prepare the piping and electrical setup for installing the Atlantium Hydro Optic Water Treatment System. If you have any questions, consult with an Atlantium application engineer.

3.1 Planning

Consider where to locate the unit from the perspective of your process. Prepare your installation plan according to the planning points below, as well as the other guidelines in this manual.

3.1.1 Access

- **Premises Access**
  - Take into account that sufficient door space is required to transport the Atlantium Unit into the facility where it is to be placed.
- **Service Access**
  - Take into consideration that sufficient space is required for maintenance personal to access the Atlantium Unit for periodic component replacement.

3.1.2 Hydraulics

- **Piping**
  - Align the upstream and downstream piping with the designated location for the Atlantium Unit.
  - Prepare the flanges/ferrules and piping as specified according to ***The Piping Infrastructure*** on page **24**. To avoid a step in the connection, make sure that the internal diameters where the Atlantium Unit and your piping match.
  - **Do not** use sealing material that contains graphite when sealing the flanges/ferrules and gaskets as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- **Mechanics**
  - Prepare the supports needed for the Atlantium Unit, noting the minimum level above the floor is at least 750mm / 29.5 inch.
  - Check that the piping on the inlet and outlet sides has adequate support structures so that there are no mechanical stresses that may endanger the Unit.
  - Check the upstream and downstream piping for vibrations. Make sure that the piping to be attached to the Atlantium Unit is anchored and reinforced to protect the Unit against vibration. A high vibration level can cause damage to any hydraulic system, including the Atlantium unit. High vibration can be caused by an unbalanced pump or by sudden valve close.
- **Hydrodynamics**
  - Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. Make sure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check your facility system procedures, including:
    - When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
    - Sudden valve closures can cause water hammer. Make sure that valves close gradually, enough to avoid this problem.
    - Ensure that all pumps that impact the Atlantium system employ soft start-up procedures.

- The Atlantium Unit must be totally filled with water. Assure that no air can be trapped in the unit and that air can be released.

3.1.3 Flow Data

- Evaluate your flow trends to determine whether you require a flow meter or flow switch. If the flow varies, a **flow meter** is required to detect and report on the flow coming into the unit. Make sure that there are to be **no** branches on or off the pipes between the flow meter and the Unit inlet. If the flow rate is static, a **flow switch** can be used instead.

3.1.4 Application

- Verify that the dose-related parameters, such as the UVT and the flow, are the same as those defined in the Atlantium application scope of supply.
- Determine what a low flow situation means to your process and whether you need to shut down the Unit (low flow indicates upstream problems) or recirculate (low flow normal) when the flow goes below the normal range.

3.2 Typical Installation Set-Up

Use the diagram below as a basis for your installation drawings. Be sure to include service access as you fit the Atlantium system into your facility (i.e., nearby pipes, electrical fixtures, support beams, etc.)

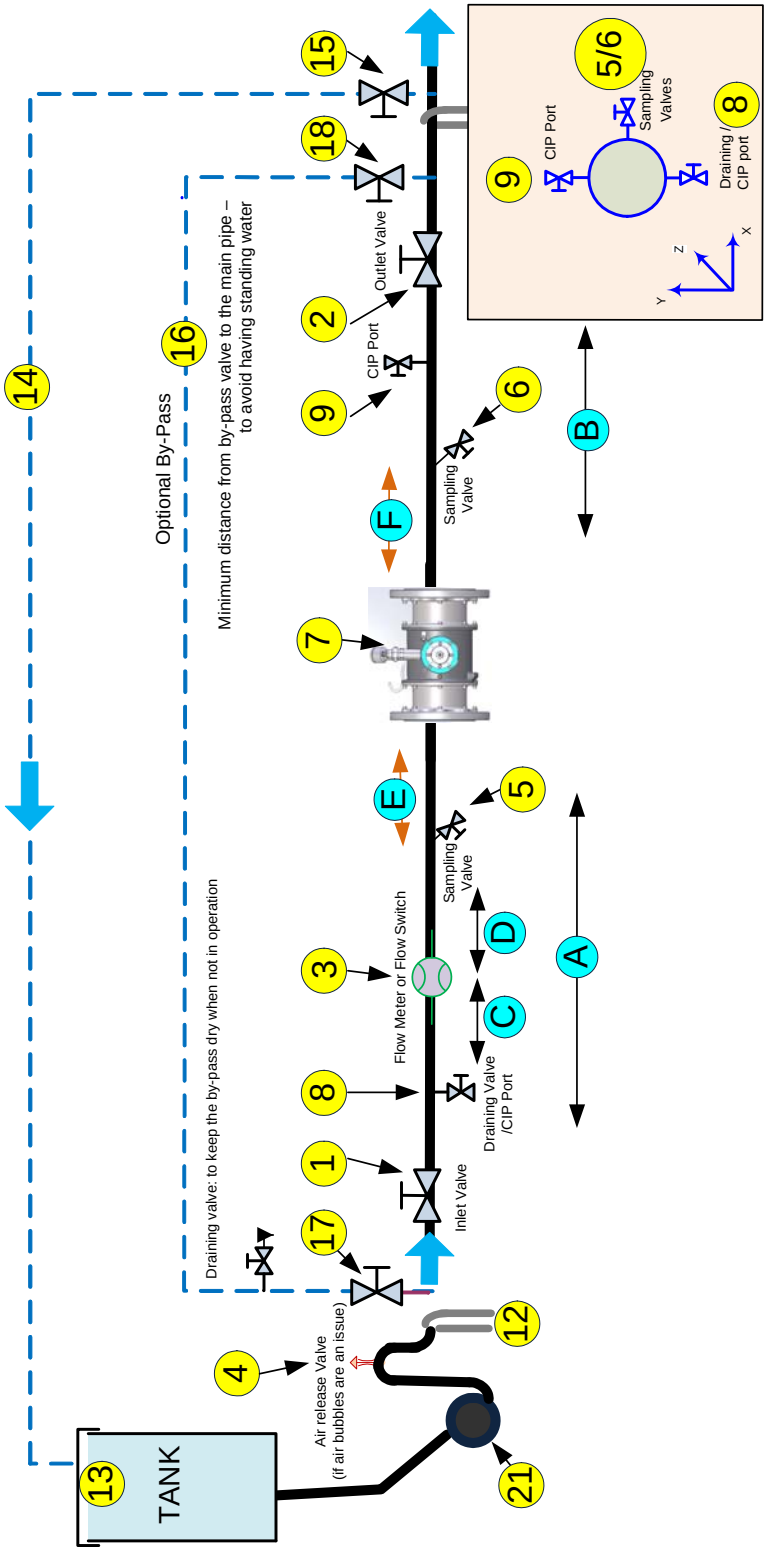


Figure 3-1: Typical Atlantium RS104 System Setup

Table 3-1: Typical Atlantium System Component Setup Key

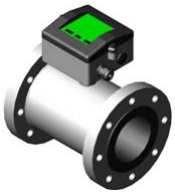
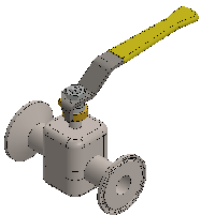
#	Description		Atlantium P/N or recommended and available at Atlantium
	Direction of water flow in the pipes		-
1	Inlet isolation Valve ■ Isolating Valve on the Inlet side ■ Stainless Steel aseptic valve		-
2	Outlet isolation Valve ■ Isolating Valve on the Outlet side ■ Stainless Steel aseptic valve		-
3	■ Flow Meter flow measurement device with 4-20mA analogue output signal ■ To be installed on the same line as the Atlantium system, with no branching before or after the flow meter ■ Follow the manufacturer's Installation instructions		Available Flow Meters: KROHNE OPTIFLUX 6000 KROHNE OPTIFLUX 2300
4	Air Release valve - To prevent air bubbles where necessary ■ Automatic Air Release valve is used to release air bubbles trapped within the line ■ Install at a high position to allow air bubbles to accumulate and release from this point		-
5/6	Inlet/Outlet Sampling Points ■ Position aseptic sampling valves **horizontally** (90^0) on the pipe center, within 50cm of the Unit ■ Aseptic sampling valve must be a dedicated valve for sampling only and resistant to alcohol and flame		■ Aseptic Sampling Valve Kit for Stainless Steel Pipe - KTB005700 ■ Aseptic Sampling Valve Kit for Plastic Pipe - KTB0011500
7	Generic Atlantium Unit (shown with flange interface. Systems are also supplied with ferrule interface.) (RZ104-11 is shown. The piping and placing of components are similar for RZ104-12.)		-
8	Draining/CIP Port - For performing CIP on the Atlantium Unit ■ The Draining port should be positioned close to the Inlet valve and between that and the Unit, ***on the bottom*** of the pipe ■ It is also used as one of the CIP ports and connects to a hose		■ CIP Kit - SAB012900 ■ CIP Valve Kit for Stainless Steel - KT0011600 ■ CIP Valve Kit for Plastic - KT0011700 (the valve itself is not part of the kits) See ***CIP Ports and CIP Kits*** on page **27.**, for information on CIP equipment.
9	CIP Port - For performing CIP on the Atlantium Unit ■ The CIP port should be positioned between the Unit and the Outlet valve, ***on top*** of the pipe, at the ***highest*** point before the isolation valve, in order to allow the pipes to be completely filled with solution		

Table 3-1: Typical Atlantium System Component Setup Key

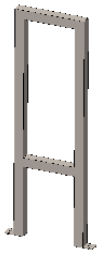
#	Description		Atlantium P/N or recommended and available at Atlantium
11	Support brackets ■ Brackets are used to hold the Atlantium Unit and pipes attached to the support mounts ■ Is to include internal flexible material to absorb vibrations and assure that the pipes and Atlantium Unit are mounted tightly and securely		-
12	Pipe Supports ■ Pipe Supports are required to assure no vibrations in the pipeline		-
13	Water tank		-
14	Recirculation Line >5m3/h (>20gpm (Optional) ■ A Recirculation line is highly recommended to keep water flowing constantly, thereby assure a high disinfection level		-
15	Recirculation Line Isolation valve (Optional) ■ Stainless Steel aseptic valve		-
16	By-Pass (Optional) ■ If the facility uses quartz corrosive CIP chemicals, such as caustic soda, use a By-Pass line to prevent damage to the Atlantium Unit		-
17	By-Pass Inlet isolation Valve (Optional) ■ Stainless Steel aseptic valve		-
18	By-Pass Outlet isolation Valve (Optional) ■ Stainless Steel aseptic valve		-

Table 3-1: Typical Atlantium System Component Setup Key

#	Description		Atlantium P/N or recommended and available at Atlantium
19	By-pass draining port (Optional) ■ The Draining port should be positioned on the bypass line, at the lowest point of the pipe		-
20	Drain ■ It is recommended to install a drain close to the draining valve		-
21	Pump ■ To prevent water hammer, the pump must be able to start up gradually.		-

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
A	Minimum 10D (1040mm/ 40inch) of straight horizontal pipe between the Inlet isolation valve and the Atlantium Unit. At least 1m / 4 inches of pipe on both sides of the ATlantium Unit must be stainless steel. (For sites with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
B	Minimum 5D (520mm/20.inch) of straight horizontal pipe between the Atlantium Unit and the Outlet isolation valve (For sites with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.)
C	Position the Flow Meter or Flow Switch typically requires a minimum 5D (520mm/20.inch) of straight horizontal pipe immediately preceding it on the Inlet side. Refer to the manufacturer's instructions.
D	Position the Flow Meter or Flow Switch typically requires a minimum 2D (208mm/11.02inch) of straight horizontal pipe immediately following it before the Atlantium Unit. A Flow Meter/Flow Switch with submerged parts is to be located a minimum of 10D (1040mm/ 40inch) of straight horizontal pipe immediately preceding the Atlantium Unit on the Inlet side. Refer to the manufacturer's instructions.
E	Position the Inlet Sampling Point Maximum Distance from the Atlantium System Inlet .5 meters/1.5ft.
F	Position the Outlet Sampling Point Maximum Distance from the Atlantium System Outlet .5 meters/1.5ft.

3.3 Electrical Requirements

The electrical infrastructure is to be installed and in place prior to the system's delivery.

For information on the footprint of the components of the Atlantium system, see the section, ***About the RS104 Systems*** on page 11. Use the diagrams below to help you plan the electrical infrastructure. See also ***Electrical Usage*** on page 13.

Take into account:

- Evaluate the quality of your power and the likelihood of blackout/brownouts, electrical surges or interruptions. Determine if you need power surge protection and or uninterrupted power supply if any.
- To verify that you order the correct lengths for cables (i.e., Cable Harness, power cables, data cable, etc.), measure the distance between the proposed location of the relevant Atlantium components and between them and the Power source.
  - Planning the placement of the Electrical Cabinet and the path of the various cables must take into consideration the length of the **Cable Harness** and the other cables. Cables must be placed in a path that does not interfere with passage ways.
  - The power mains connections must be in the vicinity of the Electrical Cabinet.
  - The power cable and connecting plug are to be obtained locally.
- If your region experience periodic power outages, consider including uninterrupted surge protection devices.
- Prepare cables of sufficient length to connect from the auxiliary equipment (i.e., the flow meter/flow switch, etc.) to the Atlantium system.

3.3.1 Wiring Diagrams

The diagrams below details the Electrical Cabinet wiring.

Electrical Cabinet Wiring for Locations with 400VAC

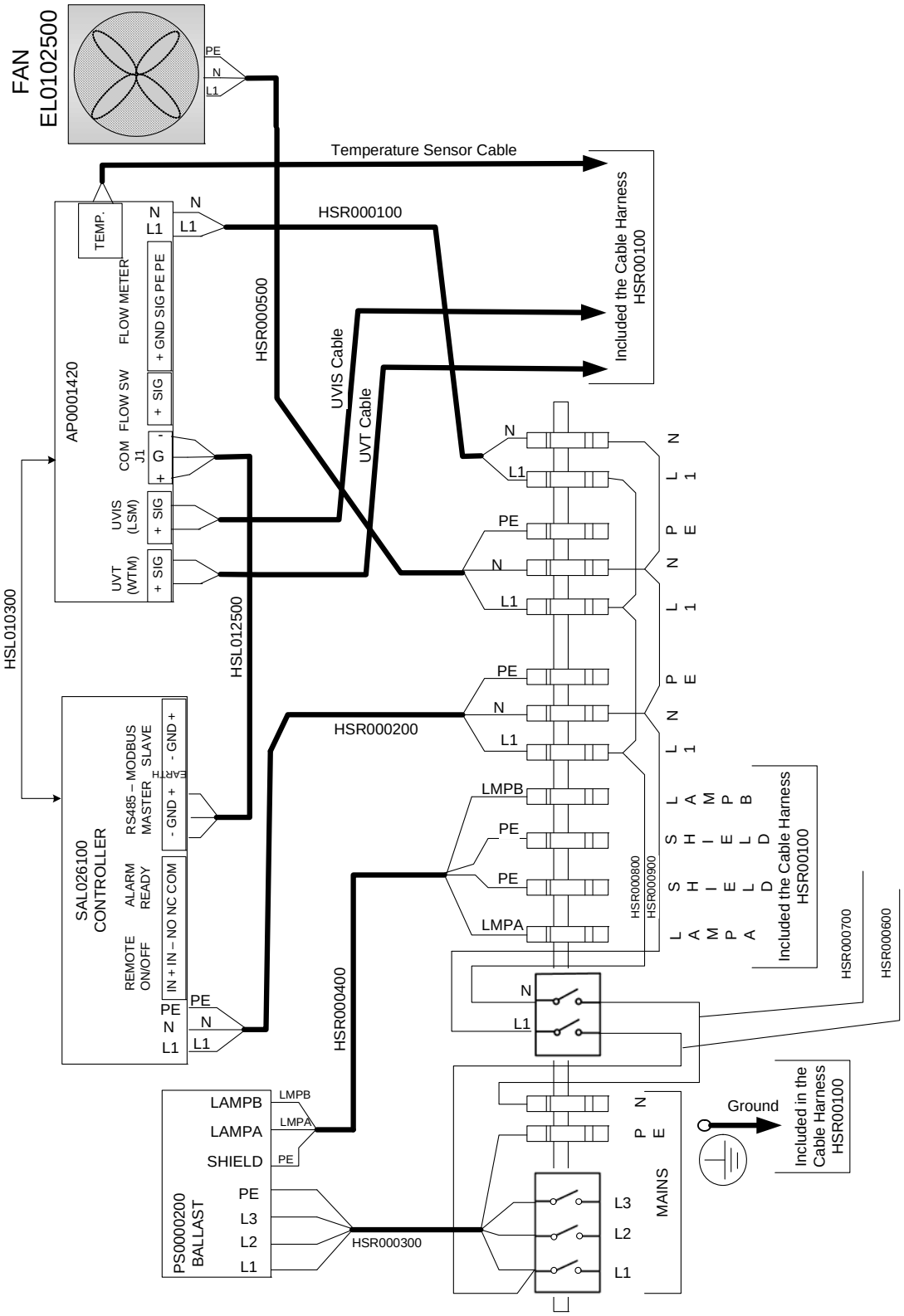


Figure 3-2: Electrical Cabinet Wiring for Locations with 400VAC

Electrical Cabinet Wiring for Locations with 480VAC

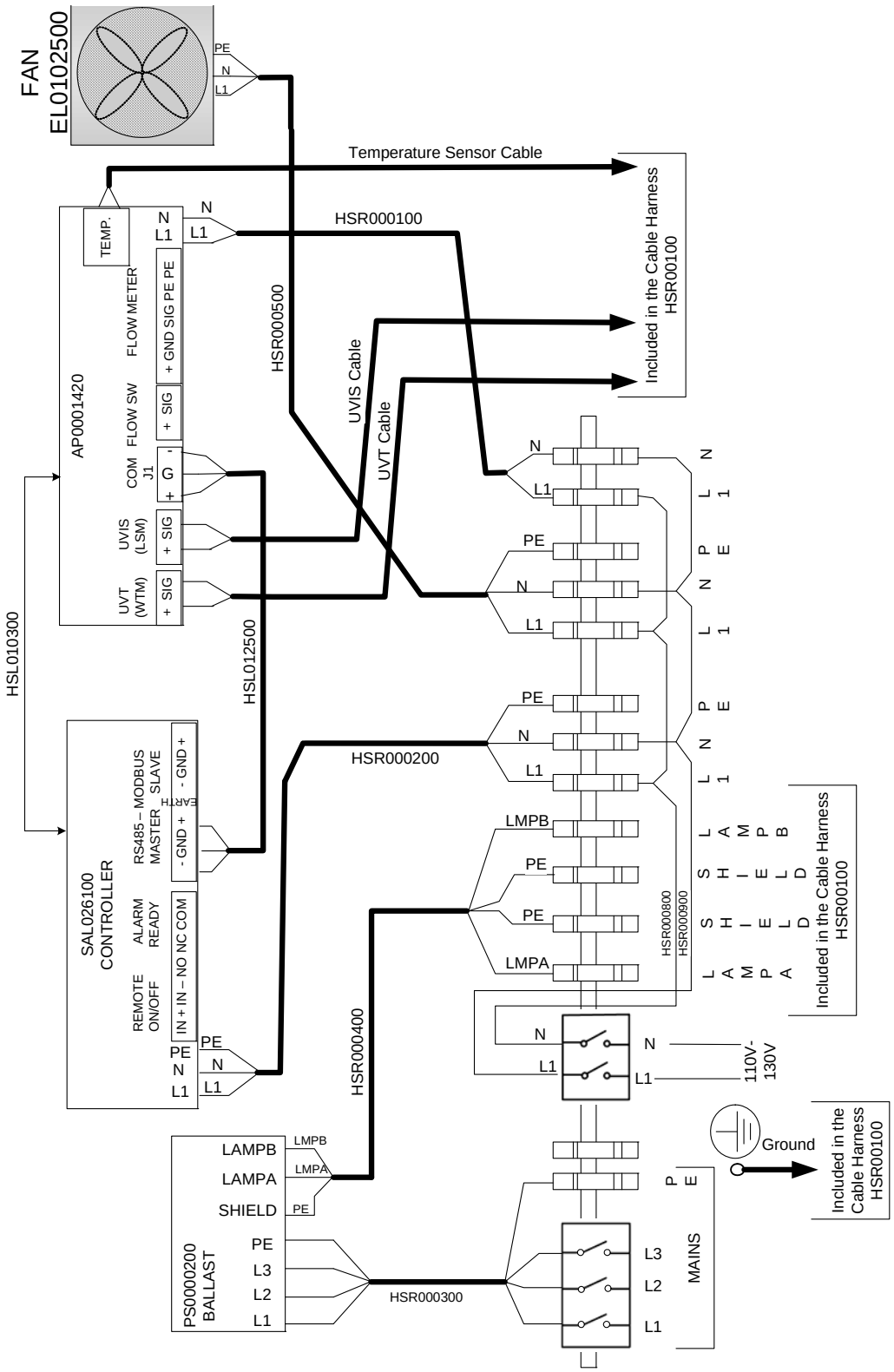


Figure 3-3: Electrical Cabinet Wiring for Locations with 480VAC

3.4 The Piping Infrastructure

To review how to prepare the infrastructure, read through the typical installation section before you begin and refer to the section, ***About the RS104 Systems*** on page 11.

Install the piping infrastructure before the Atlantium system arrives, but to avoid built-in mechanical stresses, perform the final pipe adaptations and welds when the unit is actually situated in its permanent position.

ATTENTION!

- At least 1m / 4 inches of pipe on both sides of the ATlantium Unit must be stainless steel.
- When sealing the flanges/ferrules and gaskets, do not use sealing material that contains graphite as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- Be sure to align the piping on the inlet and outlet sides of the designated location for the Atlantium Unit with the unit to avoid mechanical stress on the Unit.
- Be sure to secure the piping around the designated location for the Atlantium Unit so that there is no vibration.
- Be sure to position adequate supports on the piping line needed to hold up and stabilize the process piping on the inlet and outlet side of the proposed location of the Atlantium unit.
- Determination of support requirements and design should be made by an on-site mechanical engineer.

3.5 Flow Measurement / Detection

The Atlantium unit requires a flow signal to calculate the dose and operate properly. The flow signal is a measurement of the actual flow or simply a flow/no-flow indicator. The flow signal is collected from either a flow meter or a flow switch.

- A flow meter is required in facilities where the flow rate is variable for any reason.
- A Flow switch can be used in facilities where the flow is at a fixed, steady rate.

The Flow meter or flow switch does not necessarily have to be installed in close proximity to the Atlantium Unit. It may be more remotely positioned, provided that it is located on the inlet side between it and the Unit and there is **no** branching or splits of the water flow.

ATTENTION!

The Atlantium system cannot be operated when there is no water or no water flow. Any damage caused due to operating the system without water or without water flow invalidates the Atlantium warranty.

Flow Meter

In most cases, a flow meter with a 4-20 mA output is required to measure the flow and automatically adjust the output of the UV system. The flow meter must be able to measure the expected maximum water flow of the specified Atlantium unit.

Install the flow meter in strict accordance with the instructions supplied by its manufacturer.

Flow meters are available from Atlantium, such as:

- **KROHNE OPTIFLUX 6000** flow meter for high-purity water applications
- **KROHNE OPTIFLUX 2300** flow meter – OPTIFLUX 2000 flow meter, includes **KROHNE IFC 300** signal converter

Consult your Atlantium representative.

Figure 3-4: KROHNE OPTIFLUX Flow Meters

Flow Switch

The flow switch tells the Atlantium Unit when water is or is not flowing. The recommended flow switch is Dry Contact with minimum flow sensing of 0.2m^3/hr (1gpm).

Install it in a vertical (12:00 o'clock) orientation on top of the pipe. Follow the manufacturer's instructions.

Figure 3-5: Examples of Flow Switch

3.6 Sampling Valve Components and Kits

Sampling valve kits are available from Atlantium. The table below details them.

Sampling valve kits are available from Atlantium.

One Accessory kit is required for each Sampling point.

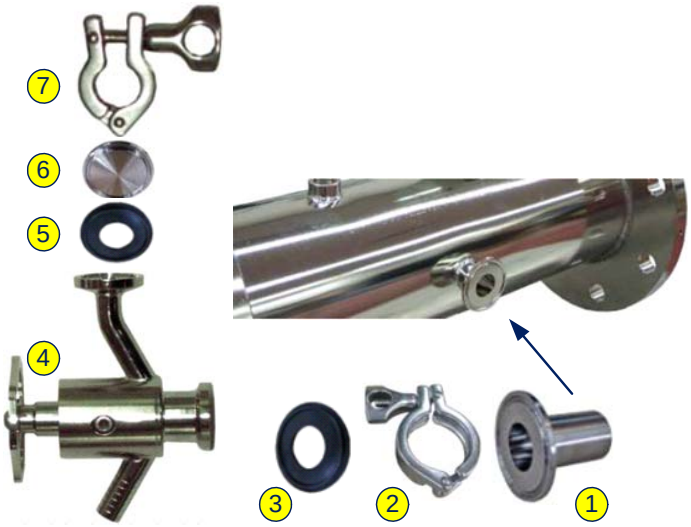


Figure 3-6: Sampling Valve Assembly

The table below details them. For information on installing the Sampling valves, see ***Installing the Inlet/Outlet Sampling Points*** on page 64.

Table 3-3: Sampling Valve Components and Kits

Description	Image	Catalog number	Kit for Stainless Steel Pipe KTB005700	Kit for Plastic Pipe KTB0011500
Long welding ferrule – for welding the tri-clamp port to the pipe		FAC003700	X	-
Threaded Male Adapter 1/2" BSP – for connecting the tri clamp port to the plastic pipe		FAB008100	-	X
Aseptic sampling valve		COB000500	X	X
Gasket 1" – for sealing the valve connected to the pipe		FA0000046	X	X
Clamp 1.5" – for connecting the valve to the welded port		FA0000045	X	X
Gasket 1/2" – for sealing the valve's flushing port		FA0000048	X	X
Clamp 1/2" – for sealing the flushing port of the valve		FA00000047	X	X
End cap 1/2" – for covering the valve's flushing port		FA0000049	X	X
Round brush – for cleaning the valve during the CIP process		FA0008300	X	X

3.7 Draining Valves

It is important to plan for draining the system. Install one draining valve at the lowest point of the pipeline, before or after the Atlantium unit. If you plan to install CIP ports, one of them can double as a draining valve. If there is a Bypass line used occasionally to bypass the Unit in any circumstance, add an additional draining valve to assure that the bypass line does become a potential source of standing water. A bypass line must be empty when not in use.

3.8 CIP Ports and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and inlet and outlet piping and sampling valves. The CIP Cart with 100 liter (26.5 gallon) reservoir and a pump, capable of pumping up to 4 cubic meters per hour. It includes a supply hose: ID 0.75" / 19mm (5 meters / 15 feet long) and a return hose: ID 0.75" / 19mm (5 meters / 15 feet long) and pocket to hold these hoses. See the diagram below.

The CIP Cart:

For systems with 400VAC: catalog number: SAB012900

For systems with 440/480VAC: catalog number: SAB012950

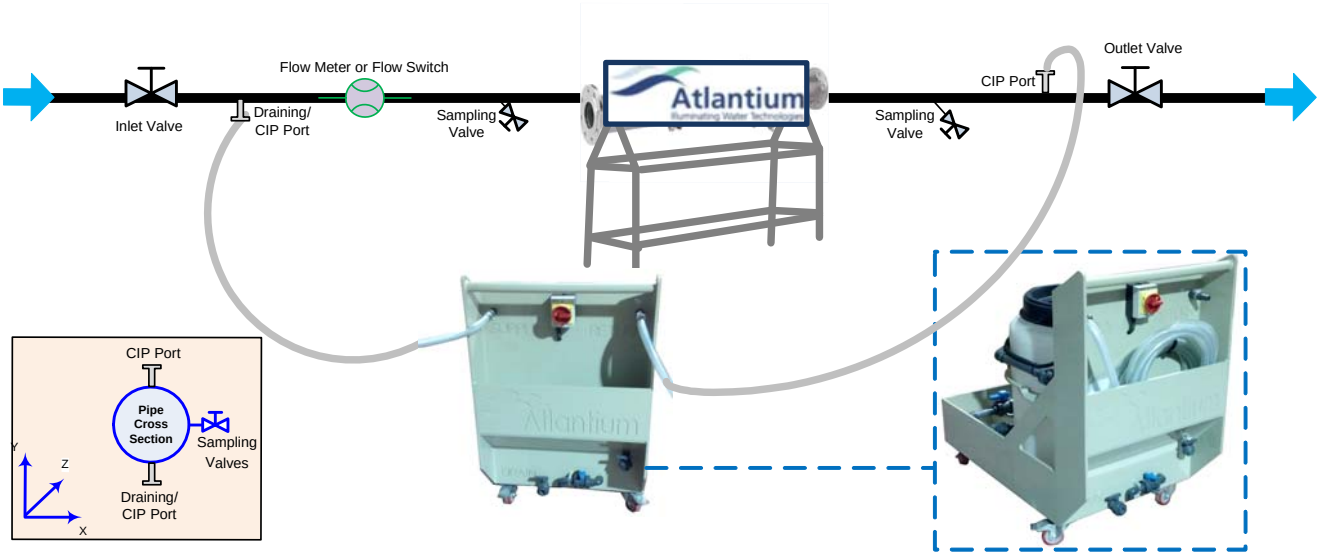


Figure 3-7: CIP Recirculation System Connections

CIP Accessories and Kits Available from Atlantium

Two Accessory kits are required, one for each CIP port.

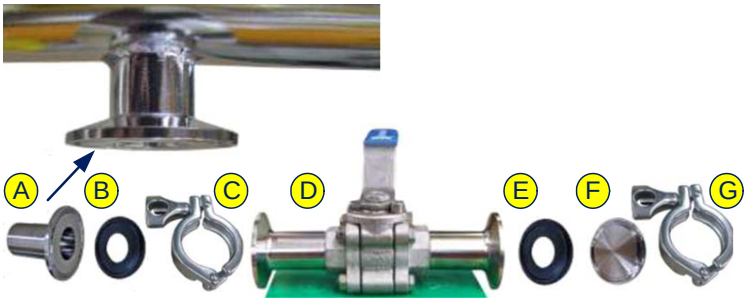


Figure 3-8: CIP Valve Kit

Table 3-4: CIP Accessory Kits

	Description	Image	Catalog number	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011700
A	**Long welding ferrule** – for welding the tri-clamp port to the pipe		FAC003700	X	-
A	**Threaded Male Adapter 1/2" BSP** – for connecting the tri clamp port to the plastic pipe		FAB008100	-	X
B,E	**Gasket 1"** – for sealing the valve connected to the pipe		FA0000046	X2	X
C,G	**Clamp 1.5"** – for connecting the valve to the welded port		FA0000045	X2	X
F	**End cap 1.5"** – covering the CIP port while not in use		FAC004100	X	X

Table 3-4: CIP Accessory Kits

	Description	Image	Catalog number	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011700
D	**Valves** - two high-purity chemical resistant stainless steal ball valve, 1 inch diameter, with tri-clamp fitting connection mechanism (A3 Standard) on both sides.			**X2** Not Included Purchased by Customer	**X2** Not Included Purchased by Customer
	Hose Adaptor 1” - to attach the CIP hose		FAB005700	**X**	**X**

CIP Safety Devices

Required but not included in Atlantium's CIP Kits: protective devices such as goggles, clothing and gloves (chemical-safe) according to the specified safety standards at your facility.

ATTENTION!

- The chemicals selected and the concentrations used must be approved for use by the facility's quality assurance and safety officials, and must comply with the written protocols and procedures set in place by those departments.
- Use protective chemical-safe goggles, clothing and gloves
- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- It is strictly forbidden to use of NaOH (Caustic Soda) as it causes irreversible damage to the quartz. Any use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

3.9 Connection to Facility Control

Options are available to set up control of the Atlantium system via your facility control system. Control setup can be made via any of the following:

- **MODBUS** - RTU on RS232 (PTP) or RS485 (network) serial connection directly to the Facility control (Appendix A, ***Modbus Communication Protocol*** on page 100).
- **Signal control** - utilizing System Ready, External On/Off, and General Alarm functions, etc. (See ***Cabling Inside the Electrical Cabinet.*** on page 41.)

With any of the above control options, you can still utilize the function of the controller of the Atlantium system. Whichever control facility is utilized, the system communicates to the other control systems such that the last command issued determines the current status.

4 Safety Overview

The Atlantium **RS104** Hydro-Optic Water Disinfection system has been designed according to the highest safety standards, assuring the safety of operating personnel, the environment and the treated water.

WARNING!

Improper use of controls or adjustments or performance of procedures other than those specified herein could result in ultraviolet (UV) light exposure and/or electrical hazards.

Therefore, personnel operating or servicing this system must be thoroughly familiar with all safety requirements and operating procedures and should adhere to them during use of this system.

To guard against injury, observe basic safety precautions, including the following:

All persons operating the Atlantium system must be aware of the potential hazards of improper use. Be certain that all personnel carefully review the safety information and the procedures specified in this manual. Only authorized individuals with appropriate training and knowledge, including local regulations, should operate, assist in the operation of or provide maintenance to the system.

All service and repair of the Atlantium system must be performed by Atlantium field service technicians or other factory-authorized personnel ONLY.

4.1 UV Lamp Safety

UV Exposure: The system generates ultraviolet (UV) light within the Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working. Do not look directly into the lamp enclosure during system operation, examining, or servicing the system's internal components during operation, or when lamps's caps are open, or if energized testing is required.

Persons potentially subject to UV light exposure – technicians who are servicing the Unit – must wear appropriate eye protection whenever the system is operating.

During maintenance procedures that are conducted with the UV lamps turned on:

- **Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

Protect Hands: **Do not touch the UV lamp with bare hands.**

Wait at least 10 minutes until the lamp is cooled down before touching the lamp.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

Lamps can reach a temperature of 1,000C° under operating conditions.

Wait ten minutes to allow the lamp to cool down fully before replacing it.

Keep materials that are sensitive to heat, or which contain solvents, a safe distance away from the lamp, Unit, and electrical connections.

Electric Shock: **Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, make sure the circuit breaker for the ballast of the relevant lamps are turned OFF and place a sign on the electrical cabinet alerting others NOT to touch the screen and circuit breakers during maintenance so that no one can inadvertently turn it on while maintenance is in progress.**

ATTENTION!

Prevent Damage: **Handle the UV lamp assembly by holding only the contact housings.**

When handling the UV lamp, always place it on a flat surface or table, so that it is not accidentally damaged.

If the body of the UV lamp is touched accidentally, clean the fingerprints off with alcohol (purity of at least 70%) and wipe dry with a soft, clean, lint-free cloth (usually provided with the lamp). Do not use cleaning rags or materials that can leave a residue.

Do not use a UV lamp that shows any scratches, cracks, or other damage.

Mercury Poisoning: **Lamps contain mercury, which is a hazardous substance. Always wear gloves and safety glasses when handling or replacing a UV lamp.**

Inhalation of vaporized mercury compounds can be harmful to the lungs, kidneys, and nervous system. Mercury that penetrates the skin or is ingested can also be harmful.

If mercury is inhaled, penetrates the skin, or is ingested accidentally, seek emergency medical treatment immediately.

If a UV lamp bursts and releases mercury, the following precautions should be observed to minimize the risk of exposure to mercury:

- Always wear gloves and safety glasses when handling or replacing a UV lamp.
- Leave the area immediately to avoid inhalation of the mercury vapor.
- Thoroughly ventilate the area for at least 30 minutes or until the mercury vapor concentration is in compliance with applicable federal and local health and safety regulations.
- After handling a broken or burst UV lamp, carefully remove gloves, and then wash hands thoroughly with soap and water. Follow all applicable federal and local health, safety, and environmental regulations.
- Do not turn on or operate the system until all mercury contamination has been cleaned up and removed.
- Make available an established Hazard Response Plan designed to deal with UV exposure and mercury contamination to all personnel associated with the Atlantium system.

ATTENTION!

Lamp Compatibility: Use original Atlantium UV lamps ONLY. **Other lamps are incompatible.**

They **void the warranty** and may damage the system.

Proper Disposal: **Disused lamps are classified as hazardous waste and must be disposed of appropriately.**

- Federal and local regulations may define how to dispose of lamps containing mercury. Atlantium cannot advise operators about their general or specific local or federal environmental and hazardous waste regulations.
- A non-ruptured UV lamp that is no longer used must be disposed of according to specific local or federal environmental and hazardous waste regulations.
- Broken lamps and parts contaminated by the mercury have to be vacuum-packed and disposed of. Lamps, broken lamps as well as contaminated packing material and other parts have to be considered as special waste. This waste may only be removed by authorized waste disposal companies.

4.2 Electrical Hazards and Safety Considerations

WARNING!!!

UNDER NO CIRCUMSTANCES
SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS Of THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE OFF POSITION AND THAT NO ELECTICAL FEED IS LIVE.
DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

- **DO NOT DRILL INTO THE ELECTRICAL CABINET OR ALTER it IN ANY WAY!!!!**
- **The Electrical Cabinet already contains entry glands for ALL of the cabling needs.**
- **Making holes in a Electrical Cabinet will destroy its waterproof seal, endangering the components inside.**
- **Do Not draw electricity from the Atlantium Unit's Electrical Cabinet as it will damage the Unit's electrical configuration.**

The Atlantium Hydro-Optic Water Disinfection system incorporates high-voltage internal components, which can cause serious injury or fatal electrical shock if not used or serviced properly. High voltage components can retain a charge for some period of time even after the system has been turned off.

- Never attempt to remove any system covers or to dismantle any parts, except when performing the maintenance procedures detailed in ***System Maintenance*** on page **10-64**.
- No portion of the Atlantium system is be opened or removed by anyone other than a trained and authorized technician.
- Do not spray or pour any type of fluid directly into the Unit or electrical components. Moisture causes damage to the equipment and electrical shock may result.
- Do not operate the system if the power cables or harnesses are frayed or otherwise damaged.

DO NOT enter **Technician Mode** in the **Controller** unless instructed to do so by Atlantium certified personnel or by an Atlantium System User Manual. for a specific task.
If you are in Technician Mode, **DO NOT WALK AWAY!**
NEVER LEAVE THE ATLANTIIUM SYSTEM WHEN IT IS OPERATING IN TECHNICIAN MODE!!
A SYSTEM LEFT IN TECHNICIAN MODE WITHOUT SUPERVISION CAN RESULT IN PERMANENT DAMAGE!!

Utilize best practices regarding electrostatic discharge (ESD).
Avoid direct contact with the electronic circuit boards of the **Controller,** which is sensitive to electrostatic discharge (ESD).
Wear an electrostatic discharge (ESD) Wrist Strap when performing procedures where contact with a circuit board is possible.

4.3 Keep the Unit Full of Water

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

4.4 Chemical Use for Cleaning in Place (CIP)

ATTENTION!

- The chemicals selected and the concentrations used must be approved by Atlantium's Application staff, and the facility's quality assurance and safety officials. The CIP process must comply with the written protocols and procedures set in place by those departments. Refer to the section, ***Selecting the Correct Chemicals for CIP*** on page **10-65**
- Use and handle all chemicals in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- NaOH (Caustic Soda) as it causes irreversible damage to quartz. It is strictly forbidden to use it without a specific analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.
- Refer the Chemical Safety information at: http://www.inchem.org/documents/icsc/icsc/eics0056.htm

4.5 System Safety Features

The Atlantium Hydro-Optic Water Disinfection system includes the following safety features:

4.5.1 UV Protective Seal

Each lamp is enclosed within a sealed lamp chamber that prevents the UV light from escaping outside the Unit during normal operations. This means that with proper operation there is no ambient UV escaping from the Unit. If, for any reason, UV light is escaping from the lamp chamber, the Unit should be serviced immediately.

4.6 Keep the Unit Full of Water

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

4.7 Software Safety Controls

ATTENTION!

When the system is started, the Atlantium software performs a series of internal self-testing routines prior to starting system operation.

If the system **does not** pass these tests satisfactorily, UV lamp operation is disabled until the problem is resolved.

If problems arise during normal operation, appropriate error messages with varying levels of severity are generated on the touch-screen control panel.

Screens that provide setup options are password protected.

WARNING!

The Technician mode screens are password protected. The Controller leaves Technician mode after a short time when inactive and returns to the main screen. However, do not leave the Controller unattended while in Technician mode even for a short time. Unauthorized changes to the Technician mode settings could damage the operation of the Atlantium system.

4.8 Atlantium Training

As part of the Commissioning and Hand-Off process, Atlantium provides hands-on training on all aspects of the Atlantium system for your designated personnel. Atlantium offers training upgrades as needed and is ready to provide additional training for new personnel or any other needs related to its systems.

5 Installing the RS104 System

Install the Atlantium Hydro-Optic UV Water Treatment System. is carried out in these main phases:

Hardware setup phases to be implemented by your facility personnel:

Phase 1 Unpacking and checking the contents of the Atlantium system packaging (See below)
Phase 2 Placing the Atlantium Unit and connecting it (See page **36**)
Phase 3 Installing the Electrical Cabinet and Atlantium Unit electrical connections (See page **40**)
Phase 4 Cabling the Electrical Cabinet (See page **41**)

Commissioning phases to be implemented by Atlantium personnel:

Phase 5 Starting up the system (See page **49**)
Phase 6 Configuring the **Controller** (See page **50**)
Phase 7 Tuning the System (See page **86**)
Phase 8 Completing the System Setup and Testing (See page **57**)
The Installation Process Flow diagram below maps out the major phases of the installation process.

5.1 Installation Process Flow Diagram

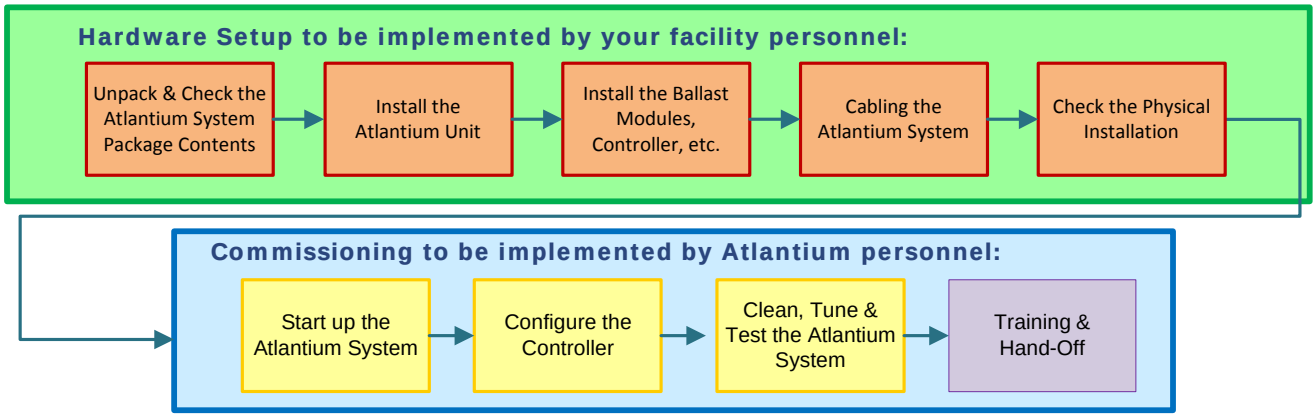


Figure 5-1: Installation Flow Diagram

5.2 Setting the Main Circuit Breakers to OFF Position

Before beginning work to connect the wires/cables of the Atlantium system components, turn the main circuit breakers of the electrical lines feeding the atlantium system components to the **OFF** position. **Verify that no electrical feed to the Atlantium system components is live.**

WARNING!!!

UNDER NO CIRCUMSTANCES SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE OFF POSITION AND THAT NO ELECTICAL FEED IS LIVE.

DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

5.3 Unpacking And Checking Package Contents

The Atlantium system is packaged and shipped in specially packed crates that contain:

- The Atlantium Unit
- The Electrical Cabinet
- Documentation
- Peripheral accessories as per your order and listed on your delivery documents are crated according to their size and weight. Small accessories may be crated together.

Each of the crates is connected to a forklift-friendly pallet. The Atlantium crates must be lifted off the transport vehicle using a forklift or crane by a qualified operator.

➲ To unload and check the system:

Step 1. Using the forklift or crane, lift each of the crates off the transport vehicle and lower them to a level ground surface.

Step 2. Walk around and inspect each crate for damage. Note if there is any separation in the joints or any other indication of any damage in transit. If there is any indication of damage, take a photograph of it and use extra caution when proceeding to the next step.

Step 3. Unclamp and remove the panels of the crates.

Step 4. As you remove the wrapping and shipping protections, verify that the contents of the crates are correct according to the delivery documents. The Atlantium Unit contains a plate with the model number and other product information. Verify that the correct product version is supplied.

5.4 Placing The Atlantium Unit

Read through this typical installation section ***before*** you begin.

Make sure to use your installation drawing during this procedure. The Unit must rest on a strong, stable, level surface and anchor the Unit.

5.4.1 Required Equipment

The following tools and equipment are required:

- Lift or transport
- Supplied bolts, washers & lock washers appropriate to your piping adaptor type
- Supports for inlet/outlet piping and for the Unit as shown on your installation plan

➲ To place the Atlantium unit:

Step 1. Transport the Unit and Electrical Cabinet to its intended location.

Step 2. At the designated locating, check the installation plan, including the minimum height of the Unit from the floor.

Step 3. Ensure that the Unit is place so that the lamps are in a diagonal position vis a vis the floor (not perpendicular to the floor).

Step 4. Carefully lower the Unit and using the appropriate bolts, washers and lock washers.

Step 5. To connect the fitting on one side of the Unit to the fittings of the facility's pipes:

- Insert an o-ring between the Unit's and the facility's pipes. In some cases, a reducer or expander may be necessary.

ATTENTION!

- When sealing the flanges and gaskets, **do not** use sealing material that contains graphite as it is corrosive to the stainless-steel components of the Atlantium system.

Step 6. On one side of the Unit to the flanged fittings of the facility's pipes with eight appropriate **(8)** sets of bolts/nuts/flat and lock washers - do not tighten the hardware.

Step 7. Repeat the step above for the other side of the Unit.

Step 8. Tighten the fastening hardware so that the torque pressure around the fittings is even, and the Unit is firmly connected to the facility's water line.

Step 9. To check for any leakage, open the Inlet and Outlet valves to allow water to flow through the newly installed Unit and its surrounding process piping.

Step 10. If you find any leakage, tighten the connections.

Step 11. Close the Inlet and Outlet valves and use the Draining valve, drain the Atlantium system piping from Inlet valve to Outlet valve.

Figure 5-2: Generic Flange fitting to the Water Line

5.5 Installing Sampling Points and CIP/Draining Ports

Both the Sampling Points and CIP/Draining Ports must be installed on the inlet and outlet sides of the Atlantium Unit.

5.5.1 Installing the Inlet/Outlet Sampling Points

Aseptic sampling valves are to be located indoors in clean environment without away from direct contact with fan or A/C air flow. They each must be a dedicated valve for sampling only and resistant to alcohol and flame. For information on the Sampling valve's components, see ***Sampling Valve Components and Kits*** on page **54**. See ***Taking Microbial Samples*** on page **6.** for information on using the Sampling valves.

On the inlet and outlet sides of the water line, connect the aseptic sampling valves,

➲ To install an inlet/outlet sampling point:

Step 1. On the Inlet side, install the **Long welding ferrule** or **threaded male adapter (1).**

- Position aseptic sampling valves **(1 & 2** in **Figure5-7**, on page **39)** **horizontally** (90^0) on the pipe center, within 50cm of the Unit.
- **For Stainless Steel. Pipes:**
  - Weld the 1 inch **ferrule (1)** clamp fitting at a horizontal orientation (3 or 9 o'clock).
  - Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
  - Passivate the pipes to avoid corrosion.
- **For Plastic Pipes:**
  - Thread the **male adapter (1)** clamp fitting to the port.
- Repeat for the Outlet side sampling point.

Figure 5-3: Generic welded ferrule (Threaded male adapter is placed in the same position vis a vis the pipe)

Step 2. Fit the 1" gasket **(2)** into the **welding ferrule** or **threaded male adapter (1).**

Step 3. Fit the aseptic **sampling valve (4)** on over the gasket **(2).**

Step 4. Fit the clamp **(3)** around the aseptic **sampling valve (4)**/gasket **(2)**/**welding ferrule** or **threaded male adapter (1)** and close and lock it.

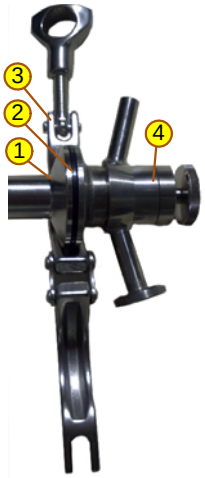


Figure 5-4: The clamp is placed around the aseptic **sampling valve** /gasket / **welding ferrule** or **threaded male adapter**

Closing Off the Aseptic Sampling Valve When Not in Use

Step 5. Into the spout of the sampling valve, fit the 1/2"gasket **(6)** and fit the cap **(7)** over it.

Step 6. Fit the 1/2" clamp **(5)** around the sampling valve spout/ gasket **(6)** /cap **(7)** and close and lock it.

Figure 5-5: The clamp is closed

Figure 5-6: The clamp is closed

5.5.2 Installing the CIP/Draining Ports

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz tube. For more information on performing CIP, see ***Cleaning In Place (CIP)*** on page **111**.

A CIP KIT includes the CIP Cart with water reservoir and a pump, capable of pumping up to 4 cubic meters per hour. It includes a supply hose: ID 0.75" / 19mm (5 meters / 15 feet long) and a return hose: ID 0.75" / 19mm (5 meters / 15 feet long) and pocket to hold these hoses. See the diagram below.

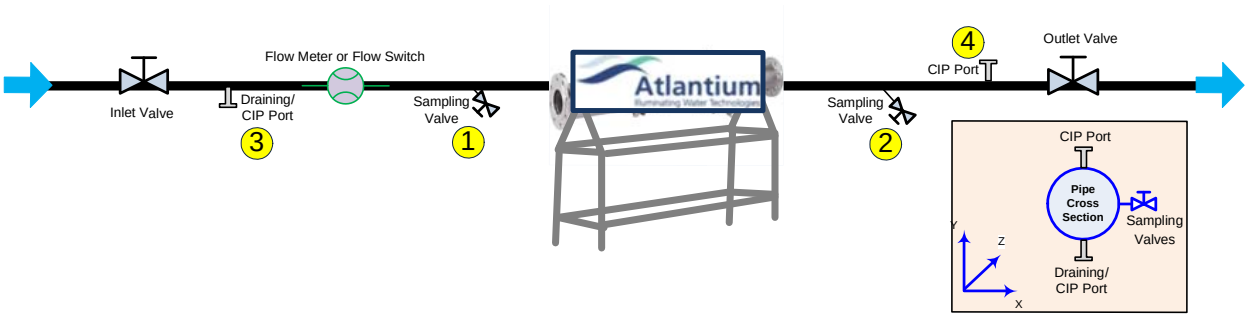


Figure 5-7: CIP Ports and Sampling Valves Orientation

Position the CIP/draining valves **(3 & 4)** on both the inlet and outlet sides of the Atlantium unit. To enable connection of the CIP kit, use high-quality, sanitary equipment, including a 1 inch tri-clamp/ferrule on both sides. A CIP/Draining valve is to be located 1D after the Inlet valve and 1D before the Outlet valve. When not using the CIP [ports. close the valve outlet using the tri-clamp cover.

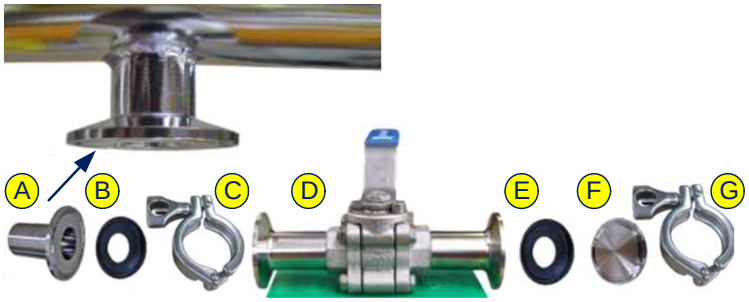


Figure 5-8: CIP Valve Kit

➲ To install the CIP ports:

Step 1. Weld or thread the 1 inch ferrule at a vertical orientation:

For municipal applications, prepare 1” screw ports for the valves and insert the valves:

- **Draining Valve:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
- **CIP Valve:** at a 12:00 o'clock orientation on other side of the Unit.

For all application except municipal:

- **For Stainless Steel. Pipes:**
  Weld the 1 inch clamp fittings:
  - **Draining port:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
  - **CIP port:** at a 12:00 o'clock orientation on other side of the Unit.
  - Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
  - Passivate the pipes to avoid corrosion.
- **For Plastic Pipes:**
  - Thread the clamp the valve fittings to the port..

Attaching the Valve to the ferrule

Step 2. Fit the gasket **(B)** into the **welding ferrule** or **threaded male adapter (A)**.

Step 3. Fit the aseptic **valve (D)** on over the gasket **(B).**

Step 4. Fit the clamp **(C)** around the aseptic **valve (D)**/gasket **(B)**/**welding ferrule** or **threaded male adapter (A)** and close and lock it.

Closing Off the CIP/draining Ports When Not in Use

Step 5. Into the ferrule, fit the gasket **(E)** and fit the cap **(F)** over it.

Step 6. Fit the clamp **(G)** around the CIP port/ gasket **(E)** /cap **(F)** and close and lock it.

Preparing the CIP Hoses

Step 7. On one end of a hose **(A)**, using an appropriate screwdriver, loosen the band **(C)**.

Step 8. Connect the hose to one of the adapters on the cart **(B)**. Push the hose up a far as you can.

Step 9. Using the screwdriver, tighten the band **(C)**.

Step 10. On the other end of a hose **(D)**, using an appropriate screwdriver, loosen the band **(F)**.

Step 11. Connect the hose to one of the hose adapters **(E)**. Push the hose up a far as you can.

Step 12. Using the screwdriver, tighten the band **(F)**.

Step 13. Repeat with the second hose to the adapter on the other side of the cart.

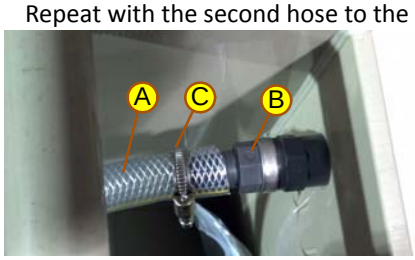


Figure 5-9: Hose Connection to One Side of Cart

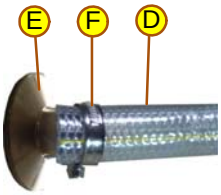


Figure 5-10: Hose Connection to Hose Adapter

5.6 Installing the Electrical Cabinet

The location of the Electrical Cabinet must be designated according to the maximum distance needed for positioning the **Cable Harness** of the Atlantium Unit. The length of a standard **Cable Harness** that connects the Atlantium Unit to its Electrical Cabinet is **5 meters (16.5 ft.)**. An optional **10 meter (33 ft.)** cable is available upon request.

Locate the Electrical Cabinet according to the design made for the path that the **Cable Harness** is to be placed.

The Electrical Cabinet can be mounted on a wall or appropriate anchoring point.

The Electrical Cabinet includes fans and filters on the top and bottom of each side. It must have at least 50cm (19.7 inches) of clearance on its right and left sides for proper ventilation. To review the complete clearance requirements of the Electrical Cabinet, see ***The Electrical Cabinet*** on page 13.

The power cable and connecting plug are to be obtained locally.

➲ To position the Electric Cabinet:

Step 1. Uncrate the Electrical Cabinet package.

The Electrical Cabinet of the Atlantium system is heavy! - It should be lifted and carried by mechanical means.

Step 2. Move the Electrical Cabinet to its location designated according to installation plan.

Step 3. Lower the Electrical Cabinet to its designated location.

Step 4. As appropriate, bolt the Electrical Cabinet to the wall or anchoring point. Refer to the diagram to the right.

Figure 5-11: RS104 Electrical Cabinet

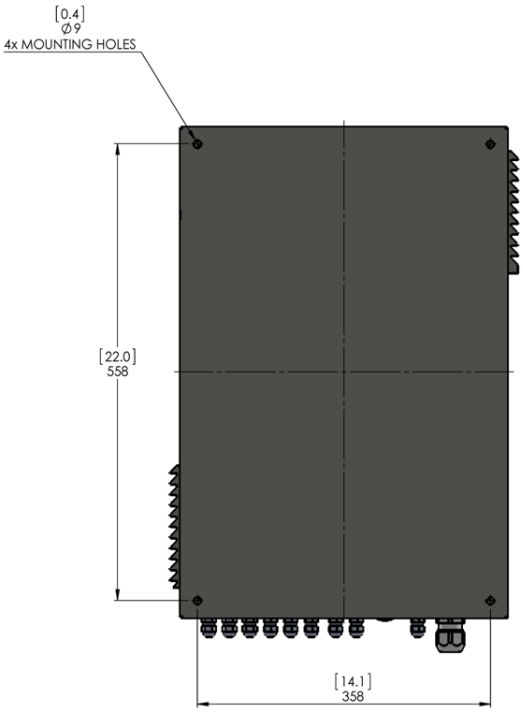


Figure 5-12: RS104 Electrical Cabinet Mounting Holes

5.7 Cabling Inside the Electrical Cabinet.

Refer to the appropriate wiring diagram in ***Wiring Diagrams*** on page **21**

ATTENTION!

DO NOT DRILL INTO THE ELECTRICAL CABINET OR ALTER THEM IN ANY WAY!!!!
The Electrical Cabinet already contains entry glands for ALL of the cabling needs.
Making holes in a Electrical Cabinet will destroy its waterproof seal, endangering the components inside.
Do not connect any wires or cables to any of the connectors on the terminal blocks or to any other connectors in the Electrical Cabinet that are not specified in your system configuration or without the express instructions from Atlantium personnel. Unauthorized connections may damage the system and void the Atlantium warranty.

Any person involved in handling the Electrical Cabinet's Connection card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.
Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the body of the Electrical Cabinet using a banana plug or an alligator clip.

Cable Harness

The Atlantium Unit comes with the **Cable Harness** assembly already connected between the **RS104** Unit and the Electrical Cabinet. You must position the **Cable Harness** assembly in a path that does not obstruct access to the Atlantium Unit or the Electrical Cabinet or any other access pathways.

In the diagram to the right, **A** identifies the Connection Card. **B** identifies the incoming cables of the **Cable Harness**.

The **Cable Harness** contains cables according to the table below.

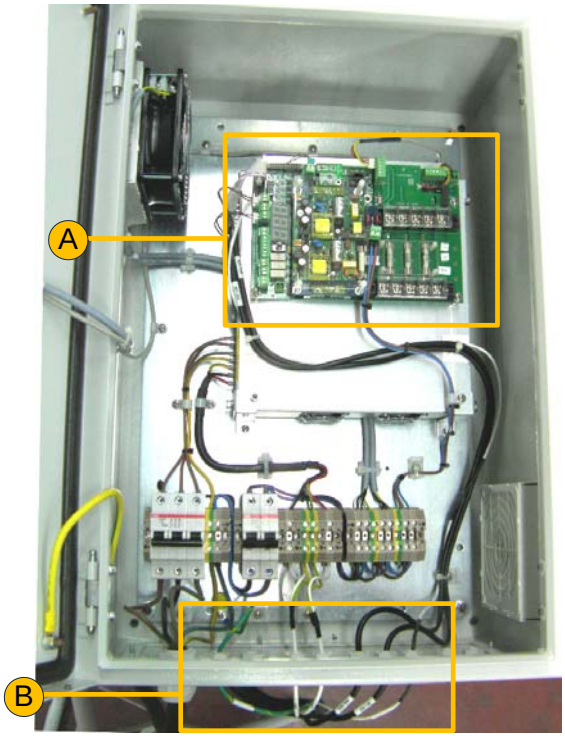


Figure 5-13: Inside the Electrical Cabinet

#	Item
1	Grounding/Earthing wire
2	110V (for locations with 480VAC only)
3	Lamp cable A
4	Lamp cable B
5	UVIS Sensor cable
6	UVT Analyzer cable
7	Temperature Sensor cable

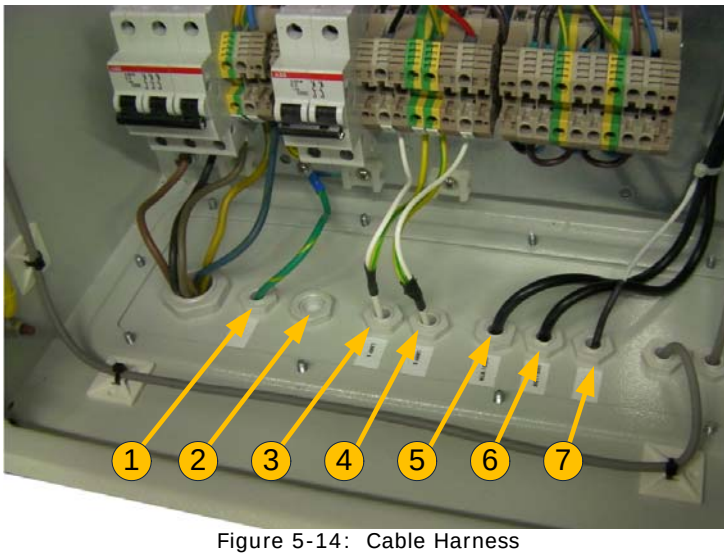


Figure 5-14: Cable Harness

Auxiliary Connections

Connect the cables of the mandatory flow signal, and External On/Off/ Alarm Ready (according to your system configuration) to the Connection card inside the Electrical Cabinet. See the diagram above for how to thread these cables and the diagram below for where to connect the cables.

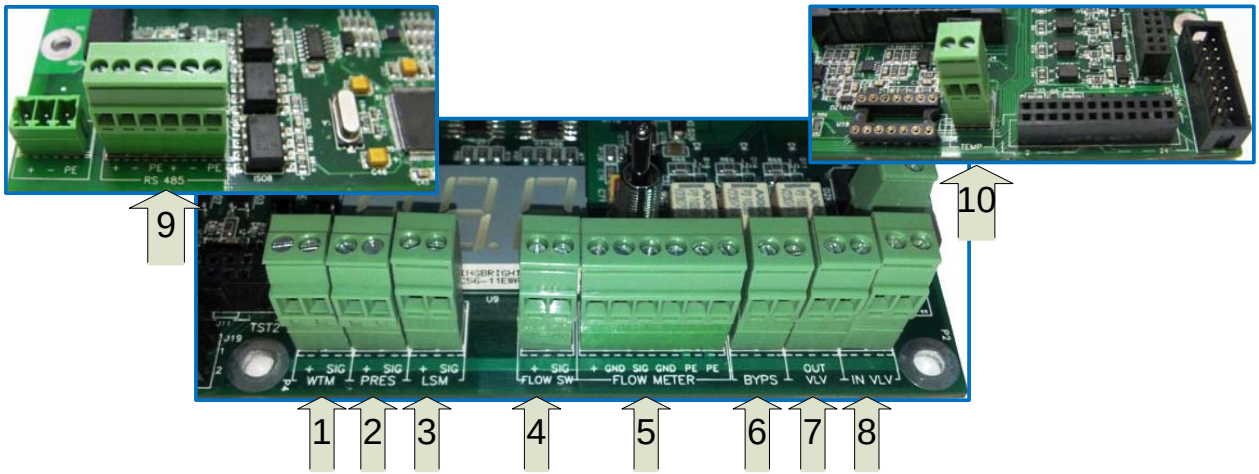


Figure 5-15: Signal Connections on Connection Card 1300

Table 5-1: Signal Connections Key (on Connection Card 1300)

#	Text on the Circuit Board	Description	Signal type	Value	Notes
1	WTM	UVT Analyzer input	Analog	4-20mA	Already assembled (Part of Cable Harness)
2	PRES	Pressure Transmitter input (Optional)	Analog	4-20mA	Optional, not implemented
3	LSM	UVIS Sensor input	Analog	4-20mA	Already assembled (Part of Cable Harness)
4	FLOW SW	Flow switch input	Digital	Dry contact / Open collector	Either Flow Switch or Flow Meter
5	FLOW METER	Flow meter input	Analog	4-20mA	
6	BYPS STR	Bypass valve output	Dry contact		Optional
7	OUT VLV	Output valve output	Dry contact		Optional
8	IN VLV	Inlet valve output	Dry contact		Optional
9	RS485	RS485 Port			Already assembled
10	TEMP	Temperature sensor input	Analog		Already assembled

➲ To connect the flow signal:

Step 1. Open the plastic gland marked **A** and thread the Flow signal cable through it until the cable reaches the Connection card terminal.

Step 2. Connect the cables to the appropriate terminal **(D)**. See **# 4** or **5** in the diagram and table above.

Step 3. Fasten the cable to the anchor point **(B)** using a plastic fastener.

Step 4. Fasten the cable to the other cables **(C)** using plastic fasteners.

Step 5. Cut the excess from the plastic fasteners.

Step 6. Tighten the plastic gland **(A).**

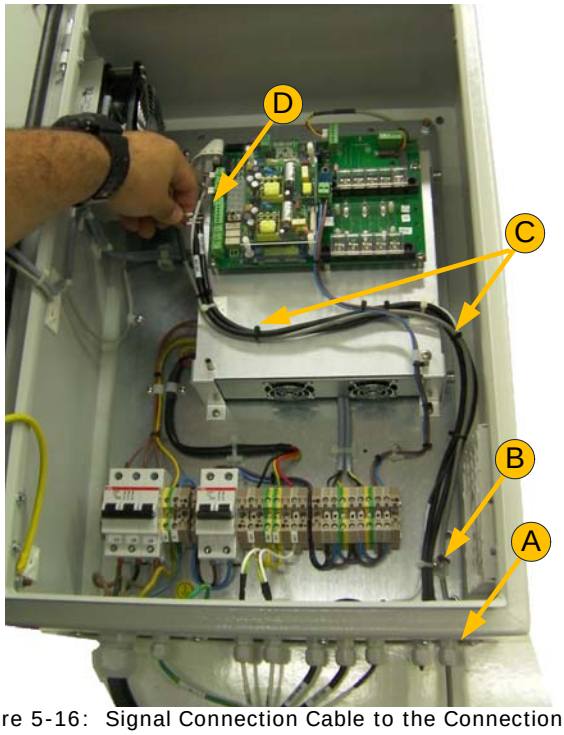


Figure 5-16: Signal Connection Cable to the Connection Card

5.7.1 Controller Connections

The Controller includes the following connections:

- Power Cable (already assembled)
- External On/Off Signal
- General Alarm Signal
- Data Cable
- Modbus

Avoid direct contact with the electronic circuit board which is sensitive to electrostatic discharge (ESD). Utilize best practices regarding ESD when replacing the power cable.

To connect peripheral equipment:

Step 1. To connect an **External On/Off** signal, thread the signal cable through one of the available holes with plastic glands on the bottom **(A)** and extend the cable **(B)** along the bottom and left side of the Electrical Cabinet, along the door until it reaches the appropriate terminal on the **Controller**.

Step 2. Using plastic fasteners, fasten the cable to the anchors **(D)** along the path.

Step 3. Connect the cable's leeds to the terminals according to the diagrams and table below.

Step 4. To connect a.n **General Alarm** signal, follow the directions as for the **External On/Off**.

Step 5. To connect to an external facility control via Modbus, follow the directions as for the **External On/Off**. For more information, see Appendix **A**, ***Modbus Communication Protocol*** on page **3**.

Step 6. Add jumpers to any unused pins.

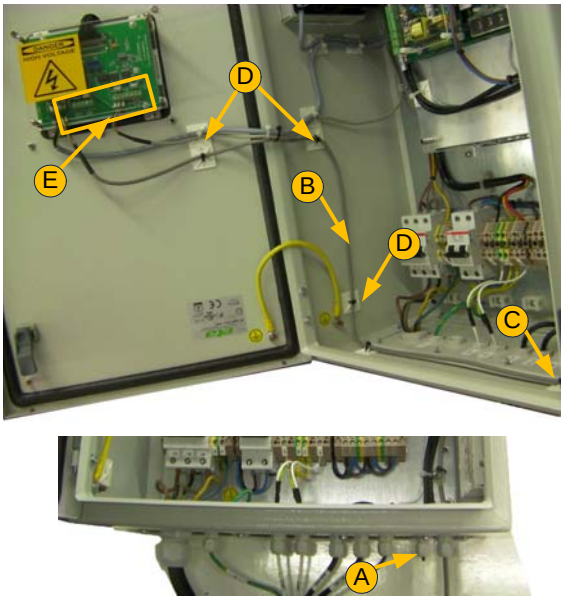


Figure 5-17: Path of Cables Added to the Controller

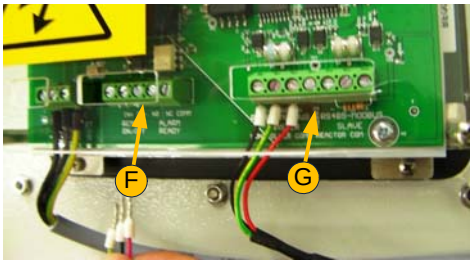


Figure 5-18: Controller Terminal Connectors-1

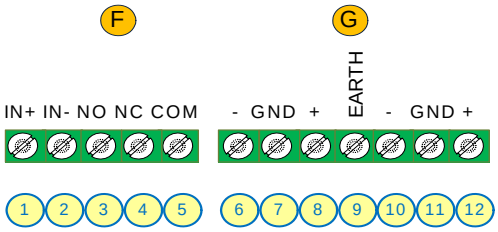


Figure 5-19: Controller Terminal Connectors-2

Table 5-2: Controller Connector Pins

#	Item	
1	External On/Off IN+	On: 20-28VDC Off: <1VDC See ***External On/Off - Additional Notes*** below.
2	External On/Off IN-	
3	General Alarm/System Ready Dry Contact Normally Open **NO**	
4	General Alarm/System Ready Dry Contact Normally Closed **NC**	
5	General Alarm/System Ready Dry Contact Common **COM**	
	Data Cable Connectors	
6	RS485 / Modbus Master TX-	Cable is provided already attached.
7	RS485 / Modbus Master Ground	
8	RS485 / Modbus Master TX+	
9	Earthing connector	

Table 5-2: Controller Connector Pins

#	Item	
10	RS485 / Modbus Slave TX- to connect to an external control system	The **Controller** is provided including a **100R, <5% , 0.25W, Through hole resistor** between **#13** and **#15**. This resistor is to be removed in **Controllers** connected by daisy-chain, but is to be left attached in the last **Controller** of the daisy-chain. See ***Cabling Inside the Electrical Cabinet.*** on page **41.**
11	RS485 / Modbus Slave Ground	
12	RS485 / Modbus Slave TX+	

5.7.1.1 External On/Off - Additional Notes

For regular wiring:

- Positive voltage must be connected to pin 4 – IN+
- Negative (GND) must be connected to pin 5 – IN-

Table 5-3: External On/Off

Controller Screen More Settings	Physical Connection	Physical Connection	
	IN+	IN-	Lamp Control
NC Unchecked	0 VDC	0 VDC	Off
NC Unchecked	+24 VDC	0 VDC	On
NC Checked	0 VDC	0 VDC	On
NC Checked	+24 VDC	0 VDC	Off

5.7.2 Setting Up the Facility System Communication Properties

To properly configure communications between your Atlantium system and control network, set up the communication properties and **COM1** port settings at the facility control system as follows:

Node Address:	2 to 254 ***
Available Commands	Function codes
Function Codes: *	0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0xF, 0x10
Support IEEE Floating Point: **	NO (register/100)
Baud Rate:	4800, 9600, 19200, 38400, 57600, 115200 ***
2 Wire	
Parity:	None
# of Stop Bits:	1
# of Data Bits	8

(*) You can read all the registers in the system at once with **0x03** (includes coils, discreet coils, holding registers and input registers).

(**) Divide the desired parameter by 100 to calculate the real flow point.

(***) Must be the same in the controller setting

To view the Modbus Registers, which govern the signalling communication from the Atlantium system to the facility control system, see Appendix A, ***Modbus Communication Protocol*** on page 100.

5.7.3 Connecting the Main Power Cable

The main power cable that you prepared for this installation must be attached to the Electrical Cabinet.

WARNING!

- The electrical connections must be performed by a qualified electrician **ONLY**.
- **Do not** plug in the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** on page 49.)

➲ To connect the power cable to the Electrical Cabinet:

Step 1. On the bottom-left side of the cabinet, open the cable gland **(A)** and insert the end of the power cable into the cabinet, ensuring that the cable can reach the main power connections **(B)** without tension.

Step 2. Fasten the cable gland's plastic nut.

Step 3. Connect the cable's power leads according to the local voltage:

- 400 VAC 5-wire system:
  - Connect three leads to the terminals marked **L1**, **L2** and **L3**.
  - Connect the Ground lead to the terminal marked **PE**.
  - Connect the Neutral lead to the terminal marked **N**.
- 440/480 VAC 4-wire system:
  - Connect three leads to the terminals marked **L1**, **L2** and **L3**.
  - Connect the Ground lead to the terminal marked **PE**

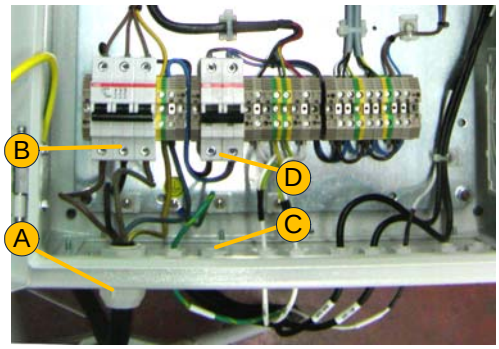


Figure 5-20: Main Power Cable Connections

The power system must be solidly grounded where the nominal line to ground voltage does not exceed 277V.
The electrical cabinet is not appropriate for use on corner grounded delta, resistance grounded and ungrounded power systems.
If the power system at the site is not appropriate you can create an appropriate power feeding by the use of 1:1 isolation transformer (by grounding the middle point of the star configuration at the secondary side of the transformer).

Step 4. **For location with 440/480VAC ONLY**, to connect the **110/120/127V** cable, on the bottom of the cabinet, open the cable gland **(C)** and insert the end of the **110/120/127V** power cable into the cabinet, ensuring that the cable can reach the **110/130V** power connections **(D)** without tension.

Step 5. Connect the leads to the circuit breaker terminals marked **L1** and **N**.

5.8 Installation Check List

Before Atlantium technical personal is to arrive to complete your Atlantium installation, Atlantium requires you review the check list below to verify that all of the prerequisites for Phase 2 of the installation are met. You must provide Atlantium with this check list at least 2 weeks in advance of the Atlantium technical personal's arrives at your site to carry out Phase 2 of the installation.

Table 5-4: Installation Check List

Item	Done
■ The System is installed: ❑ Vertically ❑ Horizontally	
■ The Atlantium Unit, Electrical Cabinet and other components are located in their final position.	❑
■ All of the System components have sufficient service clearance on all sides (according to ICDs).	❑
■ All power preparations are complete and Power connects are wired as instructed in this guide. ■ Correct voltage was measured where the Power connects to the Electrical Cabinet Voltage rate _____ Number of phases _____	❑ ❑
■ The Atlantium Unit is connected to the water pipes according to the Installation plan developed by facility engineers and Atlantium personnel.	❑
■ Water flows through the Atlantium Unit - The Unit has been checked while full of water and no bubbles or leaks exist in the system. The system is balanced, supported and pressure stable (no vibration, etc.). No known water hammers or pressure above 10 bars are on the line.	❑
■ All piping accessories are installed and wired as instructed in this guide (e.g., Inlet/Outlet valves, sampling valves, draining valves, CIP valves, air release valve, etc.) ❑ Sampling Valves ❑ CIP Ports ❑ Draining Valves ❑ Air Release Valves	❑
■ A flow meter or flow switch is installed and the electrical and data connection are complete. ❑ If flow meter: 4 - 20 mA signal and Com wires are connected as instructed in this guide. ❑ If flow switch: 2 wires from the flow switch are connected as instructed in this guide. ❑ A 4 - 20 mA signal from the PLC is installed.	❑
■ All Electrical Cabinet connections are complete.	❑
■ **You are scheduling an electrician to be on site and available during the System commissioning (a must or the installation cannot be completed).**	❑
■ To be sent to Atlantium with this checklist, attach photos of the following: ❑ Atlantium Unit ❑ Electrical Cabinets ❑ Accessories (Flow meter or flow switch, CIP ports, Sampling valves, etc.)	
Customer Approval Name of Customer's authorized representative: _____ Position: _____ Signature of Customer's authorized representative: _____ Date: _____	

6 Initializing the Atlantium System

The Atlantium system starts up with the initialization of the **Controller** on the electrical cabinet.

➲ To start up the Atlantium system:

Step 1. Check that the electrical cabinet's power cable is connected to the main power source.

Step 2. Lift all of the circuit breaker switches to the upper **(On)** position.

The system powers up and the **Controller's** touch-screen monitor is activated, displaying the **Main Operation** screen. The **Lamp Power** button on the bottom right is on the **Off** position.

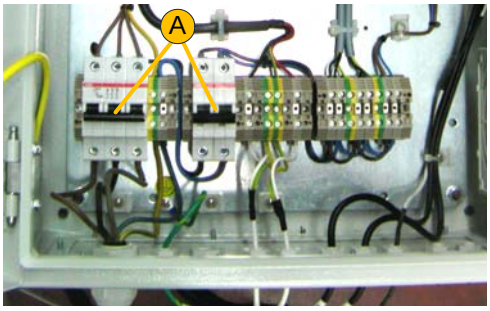


Figure 6-1: Electrical Cabinet Circuit Breakers

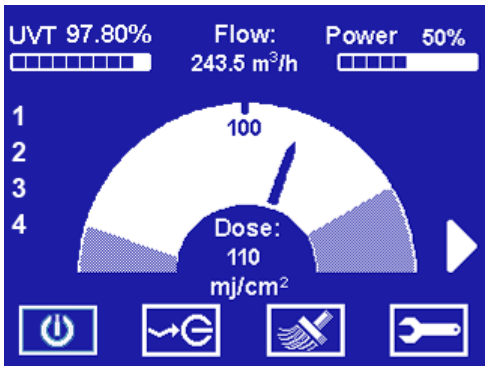


Figure 6-2: Initial Main Operation Screen

The screen is generic. The number of lamps on your Controller screen are according to the number of lamps in your Atlantium Unit.

7 Configuring the Controller

The Standard **Controller** controls the operations and dose measurements of the Atlantium system.

The System uses a validated algorithm to measure the dose and display it in real time. Depending on how you choose to run your process, you can configure the controller to maintain a specific dose or to simply use a particular power level and inform you if it does not reach the dose you designate.

The **Controller** provides a real-time display of operational data.

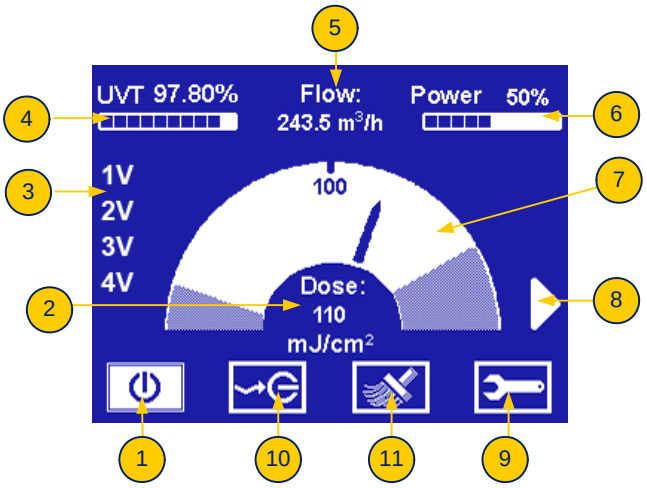


Figure 7-1: Initial Main Operation Screen (Generic View)

Table 7-1: Main screen components

#	Item
1	**Lamp Power Button** - upon Controller initiation, is on the Off position. After being pressed, tuns white, showing that the lamps are working. It can be used to shut it down in an orderly manner.
2	**Dose** - the **Op**(erate) **At d**ose is the dose the user wants the Unit to achieve and maintain. Generally, it should be set about 10-20% over the minimum dose to assure stability and on-spec operation in the case of small variations of water quality or flow. You should confer with the Atlantium application engineer in setting this number based on the capacity of the Unit, variability at the site and power considerations. When the actual dose is below the **Op**(erate) **At d**ose, the number rounds down. Otherwise, it rounds up.
3	Numbers **1-4** represents up to four lamps included in the System. **V** indicates lamps in operation. **X** indicates lamps in not operating. When the System is in an initial state, no indicator appears.
4	**UVT** - scale exhibits the current quality of the water flowing through the system, and the amount of the UV light absorbed by the water, compared to the amount UV available for disinfection. If the UVT percentage factor is high, the clarity factor of the water is high, and vice versa.
5	**Water Flow** - Number shows the current rate of water flowing through the Unit, based on a flow signal (for variable flows) or a constant volume (for on/off steady flow applications).
6	**Power** - displays the current power level. The Unit's electronic ballast varies continuously (not in limited steps) to maintain the dose and respond to changing conditions.
7	**Dose Gauge**- represents the total system dose. The gauge has two different linear functions - from the dose setting down and from the dose setting up. The minimum and maximum allowable dosage levels are user-defined.
8	Press to access the Lamps screen.

Table 7-1: Main screen components

#	Item
9	Press to access Settings and Technician protected screens via Password entry screen.
10	**External Operation button** - appears on the main operating screen only if the External On/Off feature is activated in the main Settings screen. If the feature is enabled, the system can be turned on or off from the facility's External On/Off switching mechanism. When the feature is disabled, the system may be turned off only from this screen.
11	**Cleaning System button** - appears on the main operating screen only if the cleaning feature is included in the system. Enables/disables the optionally-installed cleaning mechanism for cleaning the Unit's quartz tubes.

Plan your **Controller** configuration with your Atlantium Application Engineer to assure that the system performs properly and achieve its objectives. Review the Configuration Printout with the Atlantium Technician who is commissioning the site to make sure the settings are understood and correct. The Atlantium Technician configures the Controller. Many of its settings can be changed as experience suggests improvements in the operations and processes.

➲ To access the Settings screen:

Step 1. From the **Main Operations** screen, press . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **1357**) and press **OK**. The Settings screen appears.

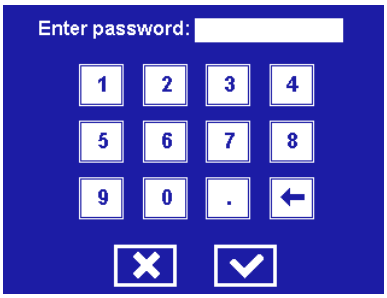


Figure 7-3: Password Screen

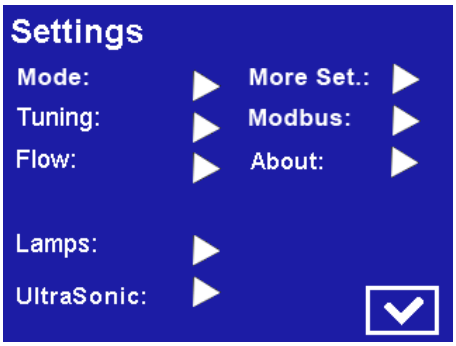


Figure 7-2: Settings Screen

7.1 Configuring the Operation Mode

The **Controller** provides Atlantium system has two operational modes. Determine which suits your objectives and facility processes: of control:

- **Power Mode** - The system is set to operate at a particular power level. Under all circumstances, the system operates at the specified designated power level. Power mode is the right choice if maintaining a particular dose is not the objective. It is often chosen where energy consumption concerns are paramount or when the maximum disinfection power is needed at all times and concern that low flows might give high doses or that high flows might give lower doses is not an issue (i.e., maintaining a particular dose is not the objective).
- **Dose Mode** - The system is set to operate at a particular dose. It maintains the designated dose as conditions allow and generally is set to alarm when the dose is not achieved.

➲ To set the Operation Mode:

Step 1. Access the **Controller's** Settings screen as detailed above and for **Mode**, press ▶. The Mode Setting screen appears

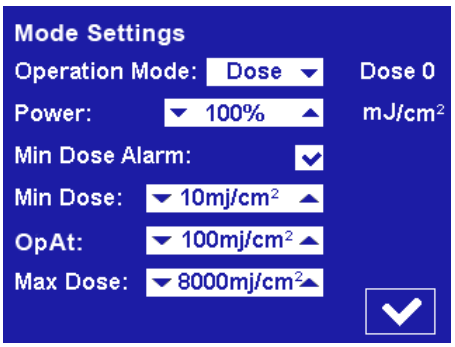


Figure 7-4: Dose Settings Screen

- **Dose mode**:
  - a For **Operation Mode**, use the ▲▼ arrows to select **Dose**.
  - b Ignore the **Power %** parameter.
  - c To enable an alarm to be triggered should the Dose reach the minimum allowed, press the **Min Dos Alarm** box until a checkmark appears.
  - d For **Min Dose**, press the ▲▼ arrows to set the lowest allowable dose in mJ/cm^2. This sets the linear function for the left side of the dose gauge and is the dose the Unit maintains. If it is not maintained, it dips into the shaded area, and, if configured to send an alarm, it does so.
  - e For **Op**(erate) **at**, press the ▲▼ arrows, to set the operating dose. This generally includes a safety factor so that the Unit does not operate at the minimum margin and does not send false alarms when the water quality or flow quivers or changes rapidly.
  - f For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge and defines when the arrow goes into the shady area. In cases where there is low flow, unusual water chemistry or the potential for overheating, high dose can indicate impending issues.
  - g Press ✓ **OK**. The settings are saved and you are returned to the Settings screen.
- **Power mode**:
  - a For **Operation Mode**, use the ▲▼ arrows to select **Pwr**.
  - b Set the **Power %** using the ▲▼ arrows.
  - c If you want to be alerted to the dose going beneath a value, set it here as the **Min Dose**. Press the ▲▼ arrows, to set the lowest acceptable dose in mJ/cm^2. This configures the left side of the gauge, but the system does not change the power or operation to achieve or maintain this dose. It is simply for information. If you set the **Minimum Dose alarm** (see below) it also triggers an alarm if the Unit falls below this dose.
  - d Ignore the **Operate at** parameter.
  - e For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge.
  - f To enable the alarm trigger if the unit goes below the minimum allowed, press the **Min Dos Alarm** box until a checkmark appears.
  - g To save your changes and return to the Settings screen, press ✓ **OK**.

7.2 Configuring the Flow Settings

The Flow signal and related settings are critical for determining the dose and assuring that the system is working within design guidelines. The accuracy and tolerance of the flow meter measurement should match the **Controller** interpretation of the flow signal.

➲ To configure the flow settings:

Step 1. Access the **Controller's** Settings screen as detailed above.

Step 2. For **Flow**, press ▶. The Flow Settings screen appears.

Step 3. For **Flow Input**, use the ▲▼ arrows to select either **Meter** or **Switch**.

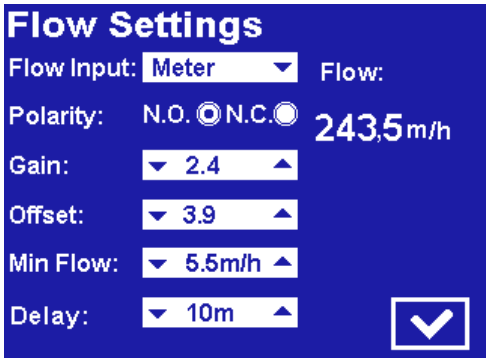


Figure 7-5: Flow Settings Screen

- If you selected **Switch**:
  - Mark the electronic polarity as **Normally Closed** or **Normally Open**.
  - The constant flow number is set by the Atlantium Technician in **Technician 2 Screen**.
- If you chose **Meter**:
  - **a** To set **Gain**, you need the meter's maximum-design flow rate found on the flow meter's datasheet. You need it in m^3/h, so if it is written in any other unit of measurement, you must convert it to m^3/h.
  - **b** In the **Gain** field, using the ▲▼ arrows, enter the meter's maximum-allowed flow rate divided by 100. (If the max flow rate is 100 m^3/h, enter the value **1**. Convert any value into m^3/h values to make this calculation.)
  - **c** To set **Offset**, turn off the flow meter and note the reading on the screen in the right hand corner. If it is not zero, using the ▲▼ arrows, enter the reading. This assures that the system registers zero in the dose calculation when the flow meter is turned off and compensates for any measurement anomalies.
  - **d** For **Min Flow**, using the ▲▼ arrows, enter the minimum flow for standard operations. Set it, making sure to convert your calculations if necessary to use the same units of measurement as are on the screen. Set this number based on process and unit considerations. The Unit cannot run without flow and in some cases, low flow may be indicative of some upstream issue. Once the Low flow number has been reached, the unit reduces power to the lowest possible power setting until the Delay has passed, and then shuts down. If the flow increases before the delay period elapses, the Unit returns to its normal operation. If the Unit shuts down, if the **Restart on Flow** is enable (see **Step 7.** on page **7-55**.), it resumes operations when normal flow is restored, otherwise it needs to be turned on manually to return to normal operations.

Step 4. For **Delay**, using the ▲▼ arrows, enter the number of minutes the system should wait before shutting down in the event of flow below the minimum.

Step 5. To save your changes and return to the Settings screen, press ☑ **OK**.

7.3 Configuring Cleaner

If Included in the system, determine how often you want to use the Cleaner and set the appropriate parameters. If not included, ignore these settings.

➲ To configure the Cleaner:

Step 1. Access the **Controller's** Settings screen as detailed above.

Step 1. For **Cleaner**, press ▶. The Cleaner Setting screen appears.

Step 2. For **Every**, using the ▲▼ arrows, set how often - in hours - the cleaning should run.

Step 3. .To save your changes and return to the Settings screen, press ✔ **OK**.

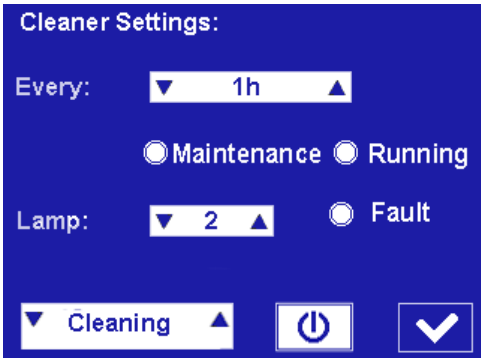


Figure 7-6: Cleaning Settings Screen

7.4 More System Settings

Access the following settings through the More Settings screen:

- Set the maximum temperature for the Atlantium unit.
- Enable/Disable Cleaning- to manually disable/enable the Cleaning system
- General Alarm - to configure a facility general alarm to be triggered when there is a system alarm or if you opt for a minimum dose alarm
- External On/Off - to configure facility control of the Atlantium system

➲ To configure more system settings:

Step 1. Access the **Controller's** Settings screen as detailed above.

Step 2. For **More System Settings**, press ▶. The System Settings screen appears.

Step 3. To set the maximum temperature threshold for the Atlantium unit, for **Max Temperature,** using the ▲▼ arrows, set the temperature.

Step 4. To manually disable the Cleaning system, press the **Cleaner En** checkbox until the checkmark is removed. To re-enable the Ultrasonic Cleaner, press the checkbox until a mark appears.

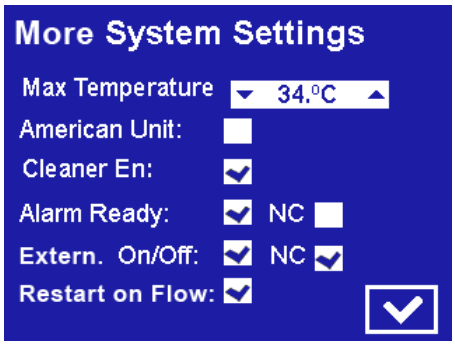


Figure 7-7: More System Settings Screen

Step 5. If your system configuration includes a facility general alarm, or system ready configuration, for **Alarm** mark the left checkbox. For **Ready**, leave the left checkbox unmarked. For electronic polarity **Normally Closed** mark the **NC** checkbox or for **Normally Open** leave the **NC** checkbox blank.

Step 6. If your system configuration includes a facility external control, mark the first checkbox.

- For electronic polarity **Normally Closed** mark the **NC** checkbox or for **Normally Open** leave the **NC** checkbox blank.
- For regular wiring, set according to the table in ***External On/Off - Additional Notes*** on page 5-46

Step 7. To set the system to restart at the configured settings when the flow returns to normal rate, press the **Restart On Flow** checkbox until a mark appears. When the checkmark is marked and the system shuts down because of low flow the main power remains on and a flow label on the main screen flashes indicating the system is waiting for a normal flow rate to resume). When normal flow rate returns, the system restarts and resumes operation according to the configured settings automatically.

Step 8. To save your changes and return to the Settings screen, press ☑ **OK**.

7.5 Configuring the Modbus

The Modbus screen displays the software version and other system version information.

Configure the Modbus node address assigned to this system in the **Controller**. The Modbus enables communication with the rest of the facility and facilitates integration of the Atlantium Unit into the overall process.

➲ To configure the Modbus:

Step 1. Access the **Controller's** Settings screen as detailed above.

Step 2. For **Modbus**, press ▶. The Modbus screen appears.

Step 3. For **Address**, using the ▲▼ arrows, set the Modbus address node to a number as per the serial address setting in your facility.

Step 4. For **Baudrate**, using the ▲▼ arrows, set the appropriate baud rate for this system. The options are:

- 115200
- 19200
- 57600
- 9600
- 38400

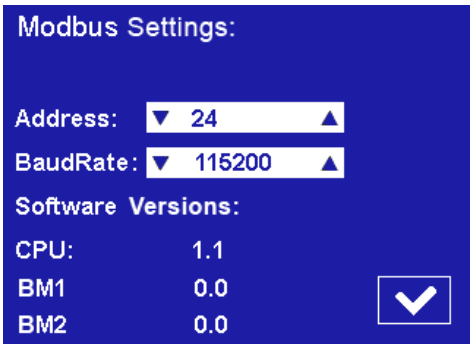


Figure 7-8: System Settings Screen

The software version and other system version information is displayed on this screen.

Step 5. To save your changes and return to the Settings screen, press ☑ **OK**.

7.6 Configuring the Technician Settings

As a precautionary measure, the Technician settings in the **Controller** should be verified.

➲ To configure the Technician Settings:

Step 1. From the **Main Operations** screen, press ⊶. The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **8888**) for the Technician Screens and press ☑ **OK**. The Technician 1 screen appears.

Step 3. For **System**, verify that the correct Atlantium system is defined.

Step 4. For **Auto On**, to set the Atlantium system to restart automatically when power is restored after a facility-wide power failure and reactivate according to the most recent settings, press the checkbox until a mark appears.

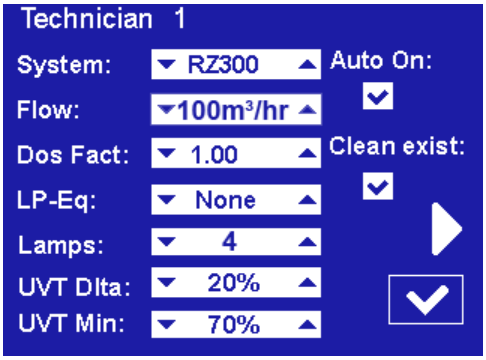


Figure 7-9: Generic Technician 1 Screen

Step 5. If the system includes the cleaning system, for **Clean exist**, press the checkbox until a mark appears.

Step 6. If the system includes a **flow meter**, for **Flow**, set the maximum flow rate using the ▲▼ arrows. Otherwise, ignore this parameter.

Step 7. If the system uses a **flow switch** instead of a **flow meter**, in the **Flow** field, set the constant maximum flow rate using the ▲▼ arrows.

ATTENTION!

Be Careful when using this screen as changing the unit time or other errors can substantially impact your system!

Step 8. If your system contains PG quartz, the Dose Factor must be configured. For **Dos Fact**, set the Dose Factor using the ▲▼ arrows.

Step 9. For facilities that require Dose monitoring in Low Pressure Equivalency terms, for **LP-Eq**, use the ▲▼ arrows to set the Low Pressure Equivalency figure. The Dose is expressed as a Low Pressure Equivalent PDM system.

Step 10. Verify that the system is configured correctly for the number of lamps. Use the ▲▼ arrows to set the correct number.

Step 11. To access the **Technician 2** screen, press the ▶ arrow to the right.

Step 12. Verify that for each lamp, the parameters listed for UVT and UVIS appear in the normal range.

Step 13. Ignore the **Power** option. It is used for testing only.

Step 14. Verify that for each lamp the column on the right is set to **ON** (enabling the lamps).

Step 15. To save your changes and return to the Settings screen, press **OK**.

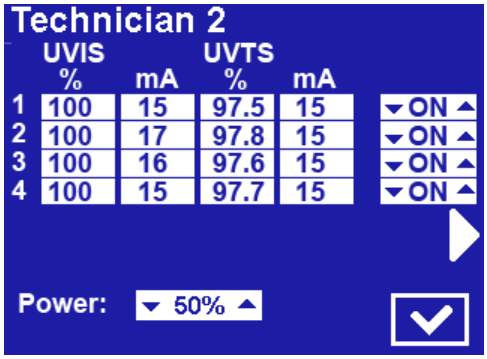


Figure 7-10: Technician 2 Screen

For a system with one lamp only, the column under #1 is displays only.

a

8 First Time System Activation

Following controller Software configuration, completing the system setup includes the following major steps:

- Cleaning the Atlantium Unit (See below)
- Introducing Water into the Unit (See below)
- Igniting the Lamps (See below)
- Initiating the Water Flow (See page 58)
- System Tuning (See page 59)
- Adjusting the Auxiliary Equipment (See page 59)
- Installation QA (See page 59)

8.1 Cleaning the Atlantium Unit

The inner surfaces of the Atlantium Unit are to be cleaned via the CIP process at installation, before starting up and proceeding with the commissioning phase. Follow the directions in ***Cleaning In Place (CIP)*** on page 65.

8.2 Introducing Water

The water is introduced into the unit before full activation.

ATTENTION!

Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. **Make sure that water hammer is NOT present in your facility's water line.**

To prevent water hammer, consult your system engineer and check your facility system procedures, including:

- When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
- Sudden valve closures can cause water hammer. Make sure that valves close gradually enough to avoid this problem.
- Ensure that all pumps that impact the Atlantium system employ soft start-up procedures.

➲ To introduce water:

- Open the **Inlet valve** and allow the RS104 Unit to fill with water completely.

8.3 Igniting the Lamps

The lamps of the Atlantium Unit are ignited to commence live system operation.

- **Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

➲ To ignite the lamps:

Step 1. **Place Caution floor signs at a distance of 1m/ 40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**

Step 2. Open each of the lamp's caps (2 on each side of a lamp). For the location of the lamps' caps, see Figure 10-10 on page 76.

Step 3. To ignite the lamps, from the **Main Operations** screen, press the **Lamp Power** button . The button changes to white and the lamps are ignited.

Step 4.

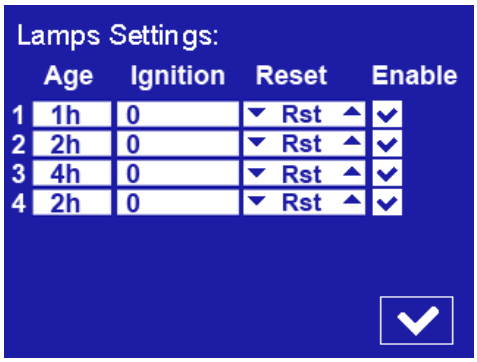


Figure 8-1: Lamp Settings Screen*

Step 5. Let the system operate for at least one hour.

Step 6. Continue with Section 8.4 and Section 8.6 below. There is no need to wait for the system to complete the first hour's operation.

Step 7. Close all of the lamps' caps.

8.4 Initiating the Water Flow

When initiating the water flow, the water may be set to drain out or collect in a tank until the system is fully operational. Once the entire installation and QA testing is complete, the water is set to flow normally as part of the facility operation.

➲ To initiate the water flow:

Step 1. Open the relevant valve that allows water to flow through the Atlantium unit. This may be the draining valve or bypass valve.

Step 1. From the **Main Operations** screen, press . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **1357**) and press **OK**. The Settings screen appears.

Step 3. For **Flow**, press ▶. The Flow Settings screen appears.

Step 4. Verify that the Flow reading on the right is within normal range. If it is not, check the connections between the Atlantium system and the Flow meter. See ***Cabling Inside the Electrical Cabinet.*** on page 41.

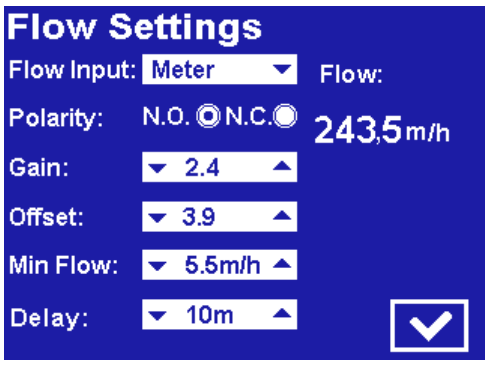


Figure 8-2: Flow Settings Screen

8.5 Adjusting the Auxiliary Equipment

All relevant auxiliary equipment must be adjusted at this time, i.e., flow meter, automatic valves, PLC, etc. Refer to the manufacturers' user documentation of the relevant equipment.

If your configuration includes a flow meter, check that the value of the flow displayed on the **Controller** matches the reading on the flow meter. See page **53** if they do not match.

8.6 System Tuning

The Atlantium system must be tuned via the **Controller**. Follow the instructions in ***System Tuning*** on page **86**.

Repeat system tuning after 100 hours of operation.

8.7 Installation QA

Atlantium personnel performs a quality assurance procedure following which the Atlantium system is ready for regular facility operation.

9 Operating the Atlantium System

This section details how to operate the system and handle basic and periodic routines to be carried out by Atlantium trained the local staff.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

9.1 Operational Guidelines

Follow these guidelines for smooth Atlantium System operations:

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

- **Incoming Water Supply** - Be alert about incoming water supply. If you do not have pre-treatment, be alert to reports of contamination by upstream slugs of mud, particles, oil, etc.

  If the water source was not previously in operation or the line is new, - flush the line prior to introducing water into the Atlantium system
- **Avoid chemicals that are harmful to UV systems** - Avoid caustic soda and other corrosive chemicals that can degrade gaskets and o-rings
- **Prevent Water Hammer** - Water hammer can cause damage to any hydraulic system, including the Atlantium unit, and steps should be taken to ensure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check you facility system procedures for following:
  - Avoid filling empty lines with water without keeping valves open for releasing the air from the pipes and the system.
  - Avoid rapid closure of water lines by sudden closure of valves.
  - Before introducing the water, ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
- **Avoid vibrations on upstream and downstream piping** - Make sure that the piping to be attached to the Atlantium Unit is reinforced to protect the Unit against vibration.
  - A high vibration level can cause damage to any hydraulic system, including the Atlantium unit.
  - High vibration can be caused by an unbalanced pump or by sudden valve close.

9.2 Basic Operational Tasks

- Shutting down the Atlantium system - See below
- Starting up the Atlantium system - See page 61
- Taking microbial samples - See page 63

9.2.1 Shutting Down the Atlantium System

You can shut down the Atlantium system from either the **Controller**. Be sure to enable the **Auto Power On** function in the **Controller** if you want the system to save your settings. (See ***Igniting the Lamps*** on page **57.**See ***Configuring the Technician Settings*** on page **55.**)

➲ To shut down the Atlantium system from the Controller:

Step 1. Open the Electrical Cabinet and set all of the circuit breaker switches to the upper **(Off)** position.

Step 1. On the Main screen of the **Controller**, press ⏻. The password screen appears.

Step 2. Using the number keys, enter the password **2468**. and press ✓ **OK.** The Atlantium system is turned off.

Step 3. Check the main screen of the **Controller** and verify that the system parameter for **Power** is **0%**

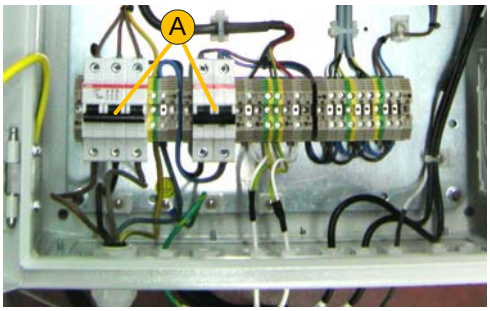


Figure 9-1: Electrical Cabinet Circuit Breakers

9.2.2 Starting up the Atlantium System

You can start up the Unit via the **Controller** and then check the water and system parameters.

➲ To start up the Atlantium system:

Step 1. Check that the electrical cabinet's power cable is connected to the main power source.

Step 2. Open the Electrical Cabinet and lift all of the circuit breaker switches to the upper **(On)** position.

Step 3. On the Main screen of the **Controller**, press ⏻

Step 4. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 5. Check the main screen of the **Controller** and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within acceptable range (see the table below to enter your expected values). Notify the executive maintenance engineer in case the values are out of range.

Table 9-1: Abbreviation of Table in Appendix B. Checking the System Parameters

	Parameter	Expected Value	
		Minimum	Maximum
1	Power		
2	Flow Rate		
3	UVT		
4	UV Dose		
5	Lamp Status		

For periodic checks on Atlantium system parameters, a convenient check list is supplied in Appendix B. Checking the System Parameters.

9.3 Taking Microbial Samples

The designated aseptic sampling valves (i.e., EGMO ESV valves) are the **Inlet** and **Outlet Sampling Points** located on the sides of the Atlantium Unit. Before taking a microbial sample, a valve is sterilized internally with Ethanol 70%, as well as with flame.

Required Equipment

- Sampling containers that are clean, laboratory-grade, sampling bottles with a volume no less than 100ml (3.38oz).
- Ethanol 100-70%
- Bunsen Burner and lighter
- Permanent marker
- Sterile gloves
- Proper Sampler attire according to local laboratory protocol

ATTENTION!

- Use gloves during sampling and treat them with Ethanol 70%.
- Use **sterile disposable** sampling containers.
- Prevent contact between the sampling containers and the valves' surfaces.
- Be sure that no foreign matter or liquids fall into the sampling containers.
- Perform the sample collection as fast as possible to minimize environmental contamination of the sample.
- **Use open flame to treat the surrounding air while sampling.**

➲ To take a microbial sampling:

Step 1. Open the Sampling valve and allow water to flow for 2-3 minutes. Then close the valve.

Step 2. If the sampling valve contains a cover at the top, open its cover and place it in a clean place.

Step 3. Spray all over the Sampling valve and inside it thoroughly with Ethanol 70%.

Step 4. Using a Bunsen burner, apply the flame comprehensively over the entire Sampling valve surface from its connection to the main pipe to its spout.

Step 5. Spray the cap thoroughly with Ethanol 70% and place it on the Sampling valve.

Step 6. Open the valve and allow the water to flow for 2 minutes. If you have applied the flame properly, vapor emerges from the spout. (If there is no vapor, application of the flame was insufficient. Repeat the previous step.)

Figure 9-2: Cleaning the Sampling Valve with Ethanol

ATTENTION!

Make sure never to touch the opening at the Sampling container's neck.
While the Sampling container is open, do not turn the cap upside down.

Step 7. On the Sampling container, with one hand, remove its cap carefully, avoiding touching the inside of the cap with fingers or any other object. **Do not turn the cap upside down.**

Step 8. With your other hand position the Sampling valve under the water stream and take a water sample. Make sure never to touch the valve's opening or the opening at the container's neck. Fill the sampling container, leaving ample air space in bottle (at least 2.5 cm).

Step 9. Carefully replace the container's cap and tighten it to avoid cross contamination.

Figure 9-3: Taking a Sample

Step 10. Labeled the Sampling container with the following data:

- Sampling site
- Sampling point
- Sampler name
- Date the sample is taken

Step 11. Close the Sampling valve.

Step 12. Spray the sampling container thoroughly with Ethanol 70%.

Step 13. Store the water samples in a dark, refrigeration unit at a temperature of 2-8oC (35.6-46.4oF) until they are examined.

9.3.1 Microbial Analysis Guidelines

The microbial analysis is to be conducted according to the company guidelines which are used in the local microbiology laboratory.

- Atlantium's recommendation is to use the microbial filtration method for the testing.
- If a number of consecutive anomalous microbiological results are obtained in one of the sampling points, take the samples only after performing CIP with a disinfection chemical (peracetic acid is recommended). It is important to let the chemicals clean and sanitize the aseptic valves, as well by letting the cleaning solution flow through them. See ***Cleaning In Place (CIP)*** on page **65.**
- In the event that system outlet counts are higher than the inlet counts, the results are to be considered as a sampling error.
- In the event that the microbial counts are not proportional in different dilutions, the results are to be considered as a laboratory error.
- For further guidance regarding microbial sampling and analysis, contact the Atlantium Application department.

10 System Maintenance

To keep the Atlantium system in peak form, preventive and periodic routine maintenance is essential. During commissioning, Atlantium trains your local staff how to do the tasks on a prescribed schedule or as needed.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

Before you begin a procedure, carefully read through it so that you can anticipate the steps efficiently.

The periodic routine maintenance schedule is:

- Weekly Maintenance Tasks:
  - Walk-Around Inspection Tour - see below
- Monthly Maintenance Tasks:
  - Checking the Ballast Cooling Fan (See page 73)
- Quarterly Maintenance Tasks:
  - Cleaning the Electrical Cabinet Air Filter (See page 87)
- Yearly Maintenance Tasks:
  - Cleaning the Quartz Tube (See page 78)
  - Replacing the Quartz Tube O-rings (See page 78)
  - Replacing the Electrical Cabinet Air Filter (See page 87)
- Maintenance Tasks Performed as Needed:
  - Cleaning In Place (CIP) (See page 65)
  - Replacing a Lamp (See page 73)
  - Checking/Replacing a Sensor (See page 82)
  - Tuning the Atlantium System (See page 86)
  - Checking/Replacing the Quartz Tube (See page 78)
  - Cleaning/Replacing a UVIS Mirror (See page 77)

10.1 Walk-Around Inspection Tour

Perform Weekly

Walk-Around and inspect the Atlantium system once a month to check for water leaks, visible damage to the cables and harnesses, or any other part of the system. Report any problem you find to an authorized technician.

➲ To perform a Walk-Around inspections:

Step 1. Walk around the Atlantium unit and inspect it. Look for leakage at the incoming and outgoing pipes.

Step 2. Inspect the **Cable Harness** and cables running between the Unit and the Electrical Cabinet for signs of fraying and any other visible damage.

Step 3. Inspect the Electrical Cabinet for any signs of damage.

Step 4. Report any leakage or damage to an authorized technician immediately.

10.2 Cleaning In Place (CIP)

The CIP procedure consists of two parts:

- Cleaning to remove the accumulated particles that collect on the quartz tube over time
- Disinfectant cleaning to sterilize the quartz tube (not applicable to some Municipal installations)

Clean the Atlantium Unit's inner surfaces and quartz tube generally on a monthly basis. However, if the system performance is on target and the UVT Sensor readings are stable, you can clean using this process less frequently, depending on the specific chemistry of your water.

A CIP recirculation system is available from Atlantium as an optional CIP Kit. Connect the recirculation system to the CIP ports to create a closed-loop flow line that pumps a diluted solution through the Unit.

The estimated time for the CIP process is 1.5 hours.

Prepare the following:

- Protective chemical-safe goggles, clothing and gloves
- An external pump-driven recirculation system with a built-in 100 liter reservoir (26.5 US gallons) or Atlantium's CIP Kit including its accessories (See ***CIP Ports and CIP Kits*** on page **27.**)
- Cleaning solution (See below)
- Disinfectant cleaning solution (not applicable to some Municipal installations) (See page **66**)

10.2.1 Selecting the Correct Chemicals for CIP

It is **critically important** to select the correct cleaning and disinfecting chemical for the CIP process. It must be powerful enough to clean the inner surfaces of the Atlantium unit, while at the same time:

- Does not harm the components of the Unit that it comes into contact with (stainless steel 316L, quartz and Viton® o-rings).
- Does not contaminate the product for which the water being treated by the system is used.

ATTENTION!

- The chemicals selected and the concentrations used must be approved for use by the facility's quality assurance and safety officials, and must comply with the written protocols and procedures set in place by those departments.

- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- It is strictly forbidden to use of NaOH (Caustic Soda) as it causes irreversible damage to the quartz. Any use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

- All chemicals must be obtained locally.
- Atlantium does not supply chemicals - get them locally from commercial chemical suppliers.

Recommended Cleaning Solutions

The following chemicals have been tested and found to be very effective for cleaning and are recommended for use when cleaning the Atlantium system:

- Phosphoric acid (H_3PO_4) 2% diluted solution
- Nitric acid (HNO_3) 2% diluted solution
- HCl 2% diluted solution - recommended for metal based (rust-colored) deposits
- Sulfamic acid (H_3NSO_3) 1.5% diluted solution - recommended for calcium based deposits

Recommended Disinfectant Solutions

(The Disinfectant CIP process is not applicable to some Municipal installations)

The following have been tested and found to be very effective for disinfection and are recommended for use when disinfecting the Atlantium system:

- Peracetic acid-based chemicals - 30 min. contact time, for example:
  - P-3 Oxonia Active 150 - 0.5% diluted solution according to manufacturer's recommendation
  - Divosan Forte - 0.5% diluted solution according to manufacturer's recommendation
- Hot water at 85°C (185°F) for 30 min.

ATTENTION!

The use of any other cleaning and disinfection chemicals requires Atlantium's approval.

DO NOT use caustic soda in the Atlantium system. If your facility uses caustic soda in a CIP process, use a by-pass line to prevent caustic soda contact with the Atlantium unit.

10.2.2 Preparing the Chemical Solution

For both cleaning and disinfecting, the liquid in the reservoir must contain the desired percentage of the chemical additive (see above). Therefore, the purchased chemical solution must be diluted and added to the reservoir. Calculate the correct amount of chemical solution to be used for the CIP process.

- The concentrations refer to room temperature.
- The diluted solution values denote the required dilution of the chemical circulating through the Atlantium Unit.
- While calculating the diluted volumes, consider all volumes including the Unit, pipes and the recirculation tank.*

➲ To calculate the correct amount of chemical solution:

Step 1. Calculate the total volume you need for the CIP* by adding the water volumes in:

- The Atlantium unit volume (measurements in liters/gallons) - Refer to the ***Physical Specifications*** Table in ***Atlantium System Footprint*** on page **12**
- The pipes between the Atlantium Unit and the **Inlet** and **Outlet Valves**
- The CIP recirculation system
- The hoses between the Unit and the recirculation system

Step 2. Calculate the amount of chemical additive to be added to the reservoir to achieve the desired dilution. Use this formula:

$$C = \frac{A * V_t}{B}$$

A - The target concentration for the liquid (chemical substance + water) in %
B - The concentration of the chemical substance to be added in %
C - Volume of the chemical substance to be added in liters/gallons
V_t - Volume of the water in the Unit + piping + CIP Reservoir (as described above*) in liters

ATTENTION!

* If you intend to collect the solution into the CIP reservoir or other container, the total volume of the cleaning solution must not exceed the volume of the CIP reservoir or container.

10.2.3 Performing the CIP Process

CAUTION!

The anti-scaling procedure is be performed only if necessary. Atlantium recommends that anti-scaling always be followed by disinfection.

The CIP process is performed utilizing fresh water. For systems with sea water, drain the system and rinse before preforming CIP.

- In facilities where CIP is performed on the plant level, circulating though all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m^3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.

This section details the CIP process with recommended steps for carrying it out. The procedure is to be adapted according to the specific circumstances of your installation.

The CIP process is to follow these major steps:

(Refer to the next section for the detailed description on the CIP process.)

Step 1. Close the inlet and outlet valves and immediately shut down the Atlantium system
Step 2. For systems with sea water, if needed, drain the system
Step 3. Remove the caps covering the CIP ports
Step 4. Connect the supply hose to the bottom CIP port and connect the return hose to the top CIP port
Step 5. Fill the CIP reservoir with fresh water or sea water
Step 6. Starting the circulation

Step 7. Adding the cleaning solution (See ***Selecting the Correct Chemicals for CIP*** on page **65.**)

Step 8. Running the CIP circulation for at least 30 minutes or as specified according to the chemical manufacturer

Step 9. Rinsing the system and the CIP cart

Step 10. Performing CIP Disinfection with the disinfection solution

Step 11. Ending CIP

The CIP diagram below is followed by the full recommend CIP procedure.

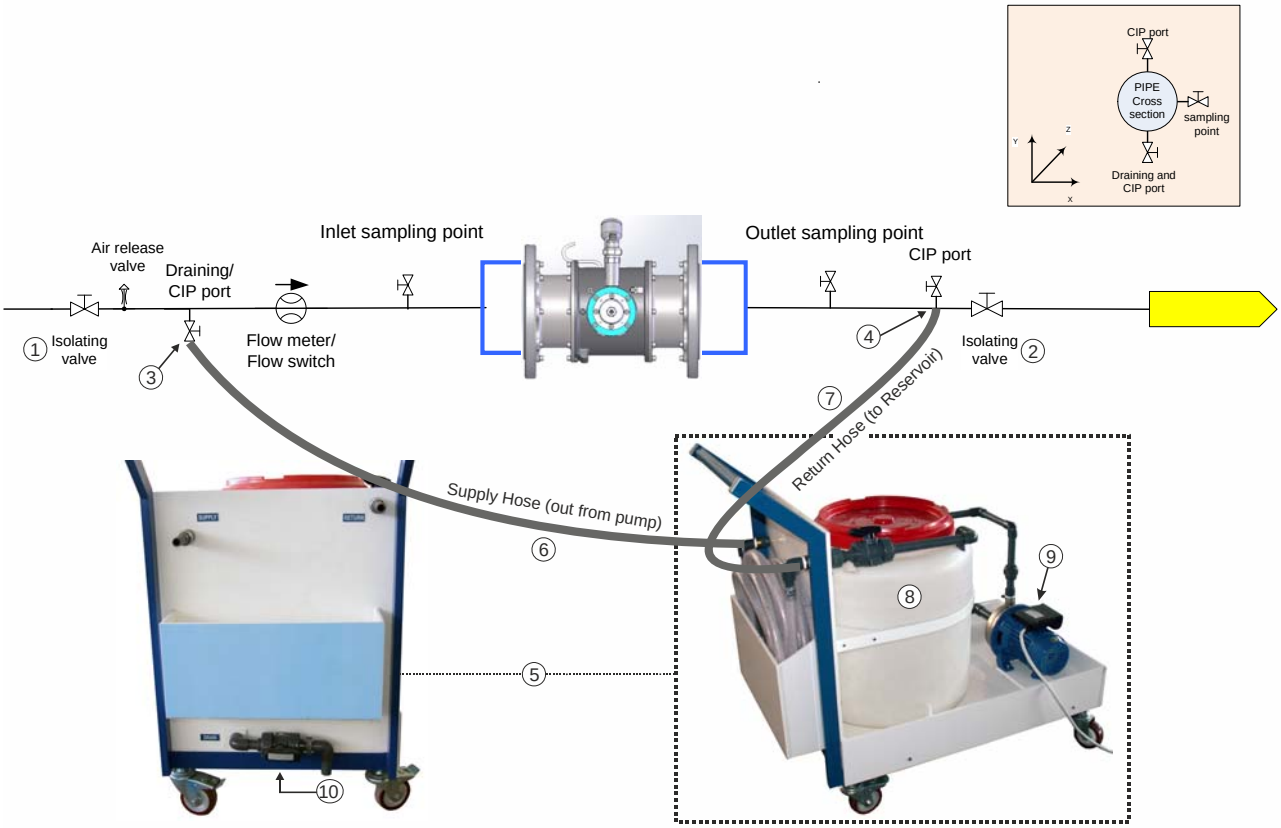


Figure 10-1: Recommended CIP Setup

➲ To Perform CIP:

If the installation is vertical, the water flow is from bottom to top. The Draining/CIP port **(3)** is on the bottom, under the Atlantium Unit and the CIP port **(4)** is above the Atlantium Unit.

➲ To Perform CIP:

ATTENTION!

If you are only performing disinfection: Follow Steps 1 through 20. Then resume with Step 22.
If you are performing both procedures: Follow all of the steps below.

Shutting Down the System

Step 1. Close the Unit's inlet isolating valve **(1)** and outlet isolating valve **(2)**. Immediately proceed to the next step.

Step 2. Open the Electrical Cabinet and set all of the circuit breaker switches to the upper **(Off)** position.

Step 1. On the Main screen of the **Controller**, press . The password screen appears.

Step 2. Using the number keys, enter the password **2468**. and press **OK.** The Atlantium system is turned off.

Step 3. Check the main screen of the **Controller** and verify that the system parameter for **Power** is **0%**

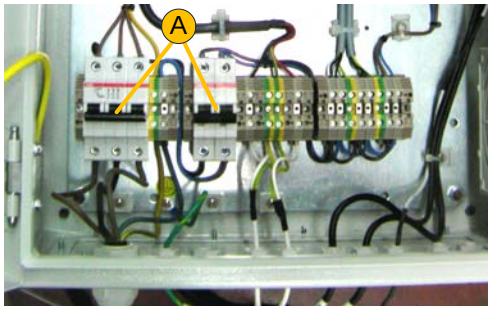


Figure 10-2: Electrical Cabinet Circuit Breakers

Step 4. Remove the caps **(D)** covering the bottom and top CIP ports **(3 & 4)**. Place the caps in a safe place for reuse later. Verify that the gasket **(B)** is still inserted into the ferrule.

Step 5. For systems with sea water, drain the system. You can use the bottom CIP/Draining port for this purpose.

ATTENTION!

Make sure that the hose fittings are secure and tight.

Water must fill the entire path through the piping on both sides of the Atlantium Unit and the Unit itself, as well as the Supply and Return hoses and the CIP Kit's reservoir.

Connecting the CIP Hoses

Step 6. Remove the hoses from their pocket on the Cart.

Step 7. Using the CIP cap's clamp, connect the CIP Kit's Supply hose to the bottom CIP port **(3)**. Verify that the gasket is present.

Step 8. Using the CIP cap's clamp, connect the CIP Kit's Return hose **(10)** to the top CIP port **(4)**. Verify that the gasket is present.

Filling the CIP Reservoir

Step 9. To fill the CIP reservoir **(A)** with water:

- **Filling option 1:**
  - Slightly open the Unit's inlet isolating valve **(1)** and open the top CIP port on the outlet side **(4)** and allow the facility's water to fill the CIP Kit's reservoir **(11)**.
  - There is a full line indicator on the reservoir **(11)**. Make sure the water in the reservoir is up to the 100 liter (26.5 US gallons) level.
    - Close the Unit's inlet isolating valve **(1)**.
    - Close the CIP port **(4)** valve.
- **Filling option 2:**
  - Open the top cover of the CIP Kit's reservoir **(11).**
  - Using a hose of an external water source, fill the reservoir with 100 liter reservoir (26.5 US gallons of room-temperature tap water (±25°C/75°F).
  - There is a full line indicator of the reservoir. Make sure the water in the CIP Kit's reservoir **(11)** is at the 100 liter reservoir (26.5 US gallons level.
  - Close the cover.

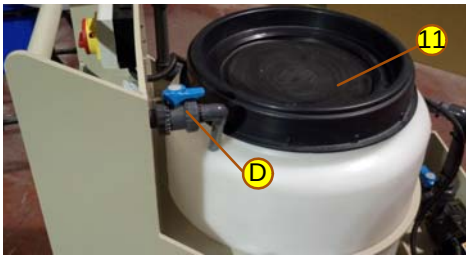


Figure 10-3: CIP Kit's Reservoir

Figure 10-4: CIP Cart's Inlet and Drain Port

WARNING!

Observe chemical safe rules! Follow all safety recommendations in the chemical manufacturer's product information and MSDS publications!

Adding the Chemical Solution

Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

Step 10. Open the top cover of the CIP Kit's reservoir **(11)** and add the prepared chemical solution to the water in the reservoir. (See ***Selecting the Correct Chemicals for CIP*** on page **65.**)

Starting the CIP Circulation Process

ATTENTION!

The Atlantium Unit must be absolutely full for the solution to reach all of the inner surfaces.

Step 11. If the bottom and top CIP ports **(3 & 4)** are valves, open them.

Step 12. To start the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **On** position and allow the solution to circulate through the Unit for **the period of time specified in the chemical's Use instructions, but not less than 30 minutes**.

Step 13. Verify that the Atlantium Unit is full of water and that the water is circulating correctly.

Step 14. During the circulation period open and close the sampling valve several times to assure chemical contact with the CIP chemicals.

If the draining valve is separate from the CIP ports, open and close the draining valve.

Rinsing the System and CIP Kit

After the chemical solution has circulated through the Unit for a sufficient duration, the chemicals solution must be rinsed from the system.

Step 15. Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system. See ***Alternatives for Draining the Water with Cleaning Solution*** on page 72.

Step 16. Open the Electrical Cabinet and pull down only the **UV PS** (**1** and **2**) circuit breaker switches to the upper **(On)** position.

Step 17. Turn on the Atlantium system, on the **Controller**, by pressing the **Main ON/OFF Operation** button in the upper-right corner.

ATTENTION!

Before performing the next step, verify that the lamp(s) are on and the system has stabilized and indicates it is ready. If not, restart the lamp(s) again and wait until the system has stabilized and indicates it is ready.

Step 18. Close the inlet CIP port **(3)**, stop the recirculation pump **(12)** and open the CIP recirculation system's drain valve **(C)**. Immediately slightly open the inlet valve **(1)** to rinse the Atlantium system thoroughly.

Allow this rinsing process to operate until at least three cycles of water volume have gone through the Atlantium Unit to remove all of the scalant cleaning/disinfectant solution.

Step 19. To assure that all the chemicals are flushed out, use a measurable indicators, such as measuring the pH, or other indicators depending on the chemical being used (consult your chemical supplier).

Step 20. Slightly open the outlet valve **(2)** and close the CIP port on the outlet side (4). To avoid hammer effect if there is air trapped in the system, carefully open the inlet valve **(1)** completely. Resume normal operation.

Rinsing the CIP Reservoir

After the chemical solution has circulated through the Unit for a sufficient duration as specified by the chemical supplier (typically 30 minutes), the chemicals solution must be rinsed from the system by the steps below.

Step 21. Open the CIP cart's drain valve **(C)** and completely drain the CIP reservoir **(11)**.

Step 22. Using an external water source, rinse the CIP cart's reservoir, letting the rinse water drain out of the CIP cart's drain valve **(C)**.

Step 23. Close the CIP cart's drain valve **(C)**.

Performing CIP Disinfection

Step 24. If you are planning to perform an additional CIP process using the disinfectant solution, refer to ***Preparing the Chemical Solution*** on page 66 and follow the CIP process again from ***Filling the CIP Reservoir*** in **Step 9.** above.

Or,

To return to normal operating status, proceed to the next step.

Ending CIP

Step 25. To drain out any water inadvertently collected in the CIP cart's storage pocket or base, use the CIP cart's drain valves **(A** and **B)**.

Step 26. Roll up the CIP Kit's hoses and return them to their storage pocket **(F)**.

Refilling the System

ATTENTION!

Make sure that the Atlantium Unit is filled with water before powering it up and turning on the lamps.

Returning Power to the Lamps

Step 27. Open the Electrical Cabinet and pull down only the **UV PS** (**1** and **2**) circuit breaker switches to the upper **(On)** position.

Step 28. Turn on the Atlantium system, on the **Controller**, by pressing the **Main ON/OFF Operation** button in the upper-right corner.

Step 29. Turn on the Atlantium system at 100% power and wait until the system has stabilized and indicates that it is ready.

Alternatives for Draining the Water with Cleaning Solution

- **If local regulations permit, and a physical drain setup exists**, drain the cleaning solution to the facility's sewage system:
  - a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
  - b With the Atlantium System operating, sightly open the Inlet valve.
  - c Close the inlet bottom CIP port **(3).**
  - d To stop the CIP recirculation, on the CIP cart, turn the switch **(13)** to the **Off** position.
  - e Open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
  - f Connect the CIP cart's drain valve **(C)** to the facility's sewage system.

Or,

- **According to local regulations, use the Draining/CIP port to drain the solution in the system into a external container.**

Or,

- **Use the CIP reservoir (11) to collect the chemical solution**.
  - a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
  - b With the Atlantium System operating, close the inlet bottom CIP port **(3).**
  - c To stop the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **Off** position.
  - d Allow the chemical solution to drain entirely into the CIP reservoir **(11)**.
  - e Dispose of the contents according to regulations. Connect a drain pipe to the CIP cart's drain pipe and open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
  - f Disconnect and close the CIP cart's drain valve **(B)**.

10.3 Checking the Cooling Fan

Once a month the cooling fan of the Electrical Cabinet must be checked to verify that the air is flowing out.

➲ To check the cooling fans:

- Check that air is being blown out of the fan in the electrical cabinet.

10.4 Replacing the Lamp

The UV lamp can be used as long as it provides the required dose and efficient service. (Generally, a lamp service status of 80% or more is considered efficient service.) A lamp should be replaced ONLY when necessary.

Each lamp's performance is measured directly by a dedicated UV Intensity Sensor. A lamp should be replaced when:

- Its status or performance declines and the lamp no longer provides the required performance at the appropriate electrical cost
- Once its rated hours expire, be alert to any changes in efficiency that could indicate the need to change the lamp

In the **Controller** e tracking application (if included), you can set a margin to assure notification as status declines and a warning is sent to the operator at both the user setting level and the internally coded minimum.

For a multiple-lamp system, the system may continue to operate while any one of the lamps is being replaced.

WARNING!

- Allow the old lamp at least 10 minutes to cool down before starting.
- Read carefully the instructions provided in the new Lamp package.
- Wear white cotton gloves.

- Before installing a new lamp, verify that it is clean and free of grease.
  If not, rub it gently with the provided lens tissue. Never touch the lamp with your bare hands - hold it only by the end-connectors. Oily residues from fingerprints can damage the lamp.

- The UV lamps are designed with high internal positive pressure. Wear protective eye-wear while replacing a lamp. Pay strict attention to the safety warnings and precautions in the ***Safety Overview*** on page **30** in the front of this manual.

- In case of lamp breakage, refer to the Safety instructions in ***Safety Overview*** on page **30**.

Generally, a lamp is replaced after shutting down the entire Atlantium system. However, in a facility where there is no redundance UV purification, and you require minimizing system downtime, you can replace a lamp while other lamps continue operation provided you follow these safety requirements:

- Shutdown the electricity supplied to the lamp to be replaced.
- **Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

Prepare the following:

- Replacement lamp
- Lamp cleaning cloth (comes with the lamp)
- White cotton gloves
- Eye protection
- #3 Allen wrench
- #4 Allen wrench
- Caution signs

To shut down the entire system:

Step 1. On the Main screen of the **Controller**, press .

Step 2. Enter the password (the default password is **2468**). The Atlantium system is turned off.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned the Electrical Cabinet's circuit breakers to the **Off** position before you proceed to replace the lamp.

Step 3. Open the Electrical Cabinet and set the circuit breaker to the **Off** position.

Step 4. Continue with the procedure, ***To replace a lamp***, below.

To replace a lamp:

To disable the lamp:

Step 1. Deactivate the lamp:

a To access the **Controller's** Settings screen, from the **Main Operations** screen, press . The **Password** screen appears.

b Using the numbered keys, enter the password (default password is **1357**) and press . The Settings screen appears.

c For **Lamps**, press ▶. The Lamps Settings screen appears.

d To disable a lamp, press its **Enable** box to remove the checkmark.

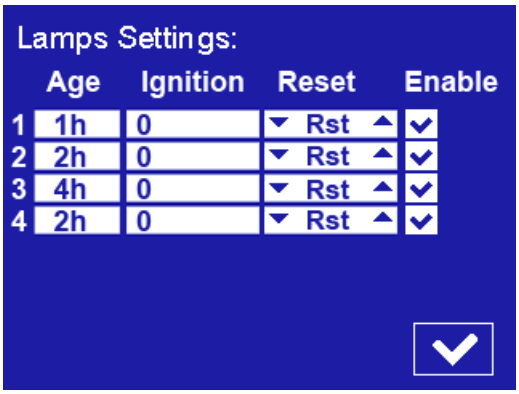


Figure 10-5: Lamp Settings Screen

If you are replacing a lamp while other lamps are still in operations, follow these safety requirements:

- **Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into the lamp enclosure during the entire lamp replacement procedure.**

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

Step 1. Allow the old lamp at least 10 minutes to cool down before starting.

Step 2. Using a #3 Allen wrench, remove the four Allen screws **(1)** that secure the lamp's connector assembly **(2)** and remove the connector assembly. Each screw has a washer.

Step 3. **To prevent any damage to the lamp, make sure you pull the lamp's connector assembly straight out and not at an angle.** Gently set the connector assembly aside on a stable surface.

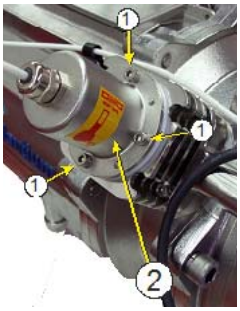


Figure 10-6: Lamp Replacement (Generic View 1)

Step 4. Using a #4 Allen wrench, remove the four Allen screws (**1**) that secure the lamp's housing (**2**), and remove the housing. Each screw has a washer.

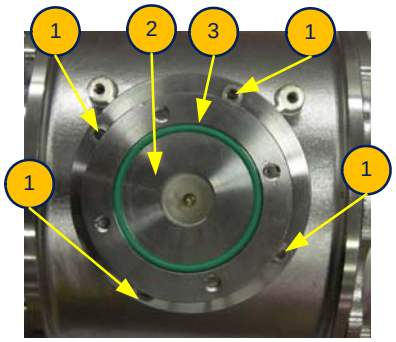


Figure 10-7: Lamp Replacement (Generic View 2)

Removing the Faulty Lamp

The lamp is hot! DO NOT touch the lamp with bare hands, particularly the quartz portion.
Wait at least ten minutes until the lamp has cooled down or use an appropriate pair of long-nose pliers to carefully pull out the lamp.

Step 5. Wearing appropriate cotton gloves and skin covering, and using a suitable long-nose pliers, pull the lamp straight out of the quartz tube in the Unit and set it aside on a stable surface.

Step 6. Check the inner surface of the quartz tube for residual moisture or high humidity. If any visible moisture exists, using a lens cleaning cloth, dry it.

Step 7. Verify that the internal sensor's mirror is cleaned.Instructions are on page **77**.

Figure 10-8: Lamp Replacement (Generic View 3)

Installing the Replacement Lamp

Step 8. Unpack the new UV lamp from its vacuumed sealed package.

Step 9. Make a note of the serial number of the new lamp.

Step 10. Clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp.

Step 11. Insert the new lamp into the quartz tube in the Unit, all the way until it stops. Ensure that the male contact point **(4)** on the end of the lamp has meshed with the female contact point**(4)** in the lamp's housing on the other side of the Unit.

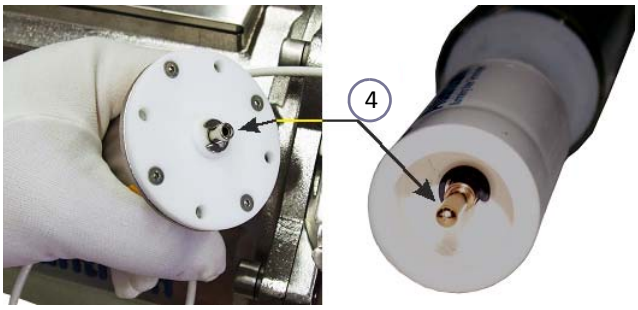


Figure 10-9: Lamp Replacement (Generic View 4)

Assembly

Step 12. Check that the lamp housing's O-ring is in place, clean, dry and undamaged. Replace if necessary.

Step 13. Insert the lamp's housing that you removed in **Step 4.** above. Ensure that the male contact point on the end of the lamp has meshed with the female contact point in the lamp's housing on the other side of the Unit.

Step 14. Replace the lamp's housing and, using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Replace the lamp's connector assembly pushing it straight in and using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Step 15. **Place Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**

Step 16. Unscrew and remove the two caps **(5)** on each side of the lamp. (See the figure to the right.)

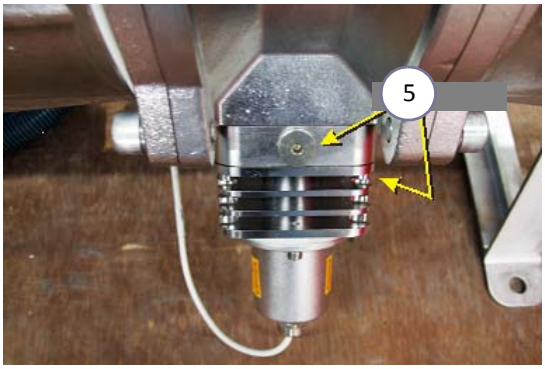


Figure 10-10: Lamp's Caps (Generic View 5)

Turning on Electricity to the Lamp

Step 17. Open the Electrical Cabinet and locate the ballast of the lamp to be replaced. Turn the corresponding circuit breaker up to the **On** position.

Step 1. Return to the **Controller** and on the Main screen of the **Controller**, press ⏻.

Step 2. Return to the **Controller's** Lamps Settings screen and activate the replaced lamp by using the ▲▼ arrows, set the desired lamp to **ENB**.

Step 3. To reset the lamps' usage clock, press ↻.

Step 4. Press ✓. The settings are saved and you are returned to the Settings screen.

Step 5. Check that the lamp ignites properly.

Step 6. Operate the system without the lamp's caps for **one hour.**

Step 7. Return the lamp's caps **(5)**.

System Tuning

Step 8. Perform system tuning according the instructions on page **86**.

Recording Lamp's Serial Number

Step 9. Gather the following information about the old lamp you replaced:

- lamp's serial number
- system model
- the number of hours on the lamp usage clock when you changed the lamp
- the number of ignitions of the lamp
- the status of the lamp when you changed it

Step 10. Email this information to:support@atlantium.com.

Observing Lamp Efficiency

Observe the system's efficiency for **100 hours**. If after this period there is a performance degradation of more than 5%, remove the UV intensity sensor and check for humidity. Clean if necessary. See the instructions on page **82**.

Step 11. Perform **System Tuning** according the instructions on page **86**.

10.5 Cleaning / Replacing the UV Intensity Sensor's Mirror

For each lamp in the Atlantium Unit, there is a small internal mirror for directing UV light into the sensor. The UV Intensity Sensor's Mirror must be cleaned every time the lamps are replaced.

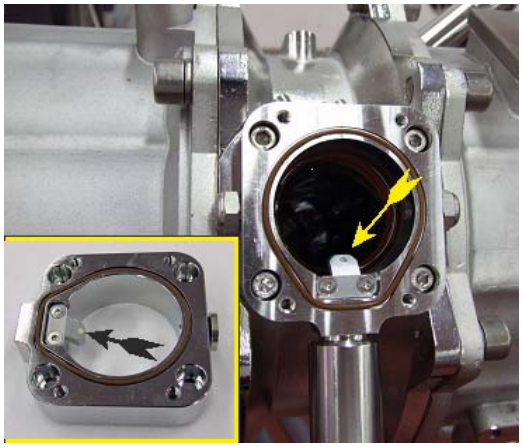


Figure 10-11: Generic Cleaning the UV Intensity Sensor's Mirror (Generic view)

Prepare the following:

- Replacement UV Intensity Sensor's Mirror
- Isopropanol and lint-free lens cleaning paper
- Appropriate Phillips screwdriver

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

➲ To clean the mirror:

Step 1. No particular parts need be removed from the Unit in order to access the mirror, once the lamp has been removed. See page 73 for instructions on replacing the lamp.

Step 2. Clean the reflective side of the mirror – the side facing into the Unit – with isopropanol and lint-free lens cleaning paper.

Step 3. Ensure that the mirror is completely dry before re-assembling the UV lamp.

Step 4. Examine the mirror's surface after cleaning and before you close the UV lamp' housing. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the mirror's surface is damaged, it must be replaced.

➲ To replace the mirror:

Step 1. Once the lamp has been removed, you can replace a damaged mirror. See page 73 for instructions on replacing the lamp.

Step 2. Using an appropriate Phillips screwdriver, remove the two screws holding the mirror in place and remove the mirror.

Step 3. Clean the reflective side of the replacement mirror with isopropanol and lint-free lens cleaning paper.

Step 4. Examine the mirror's surface after cleaning and before you install it to make sure it is damage-free.

Step 5. Carefully place the new mirror in position and using the screwdriver, fasten it securely.

Step 6. Before you close the UV lamp' housing, examine the mirror's surface it make sure it is free of fingerprints or other stains, which may cause erroneous UV intensity sensor readings. Clean it again if necessary.

Step 7. Ensure that the mirror is completely dry before re-assembling the UV lamp.

10.6 Cleaning / Replacing a Lamp's Quartz Tube and O-rings

Performed as needed

The quartz tube is cleaned as needed according to the UVT Sensor readings and when deposit collects on the quartz tube. **The quartz tube's rubber O-rings must be replaced when dismantling the quartz tube**.

Under normal environmental conditions, the quartz tube's rubber O-rings (as shown in view B in **Figure 10-12** above) must be replaced once a year even if they pass visual inspection. For environments where the water temperature is above 70°C (160°F), the quartz tube's O-rings must be replaced quarterly, for which after the first year, O-rings may require replacement less frequently. For more information, consult your Atlantium representative.

Depending on the quality of the water being treated by the Atlantium Unit, the exterior surface of the quartz tube that houses a lamp may require periodic cleaning or replacement if after the optional Ultrasonic Cleaning system and the CIP process the quartz tube is still not sufficiently clean. This is to be determined by the quality control officer at the facility.

Prepare the following:

- Replacement quartz tube (if replacing)
- Two O-rings (See view **B** in **Figure 10-12** below)
- #3 Allen wrench
- #4 Allen wrench
- Protective gloves
- Appropriate cleaning cloth. A mild, household cleaning solution
- Alcohol

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

➲ To clean or replace the quartz tube:

Powering Down the System

Step 1. On the Main screen of the **Controller**, press ⏻.

Step 2. Enter the password (the default password is **2468**). The Atlantium system is turned off.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's circuit breaker before you proceed to replace the lamp.

Step 3. Open the Electrical Cabinet and locate the main circuit breaker and turn it down to the **Off** position.

Isolating the Unit

Step 4. To isolate the Unit, close the Inlet and Outlet valves.

Step 5. Using the Drain valve, drain the water from the Unit.

Accessing the Quartz Tube

Step 6. Quartz tube removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) and the lamp be removed.Follow **Step 2.** to **Step 5.** on page **75** and remove the lamp connector assembly and lamp housing from the other side of the Unit, as well as the lamp, placing them in a secure location until they need to be reinstalled.

ATTENTION!

While pulling the lamp housing away, DO NOT tilt it and DO NOT pull it away from the lamp/sleeve axis at an angle

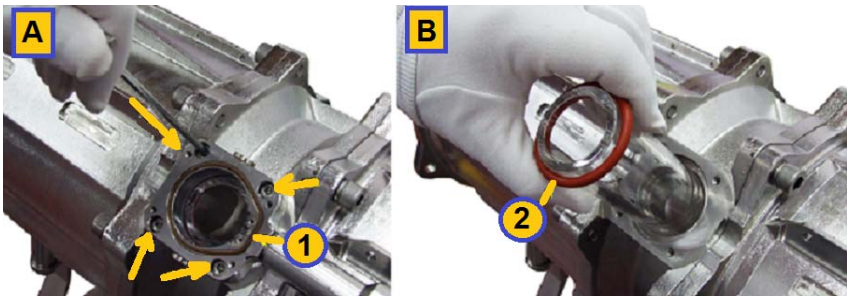


Figure 10-12: Replacing the Lamp's quartz tube (Generic View 1)

Step 7. On one side of the quarts tube, using a #4 Allen wrench, remove the four Allen screws, opening them in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) (arrow indicate the screws in view **A**) that secure the tube's holding ring to the Unit. Loosen one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the tube (2 turns).

Step 8. Continue to alternate until both tube holders are loose, and then remove the screws completely.

Step 9. Inspect the now visual O-ring (**1** in view **A**) in the Lamp housing. If there is visual damage, replace the O-ring.

Step 10. On the other side of the quarts tube, repeat **Step 6.** to **Step 9.**

ATTENTION!

Wear protective gloves. To protect the quartz tube, do not touch the internal surface with your bare hands.

Step 11. On one side of the quarts tube, place the gloved palm of you hand over the quartz tube and gently push in. The quartz tube shifts inward. The o-ring on the side where you pushed slips out (if not, take it off).

Step 12. On the other side of the quartz tube, the tube sticks out slightly. T**o prevent any damage to the lamp, make sure you pull the lamp's connector assembly straight out.** Carefully pull out the quartz tube and remove the O-ring (shown in view **B** above).

ATTENTION!

While pulling the connector assembly away, DO NOT tilt it and DO NOT pull it away at an angle.

Cleaning the Quartz Tube

Step 13. If there is a deposit buildup on the tube's exterior surface, clean it with a mild, household cleaning solution.

Step 14. Clean the exterior and interior surfaces with alcohol. If the tube cannot be cleaned to a pristine condition, replace it with a new one.

Step 15. Ensure that there is no residual moisture or high humidity on the inner surface of the quartz tube. If any visible moisture exists, dry it before inserting it in place.

Installing the Quartz Tube

Step 16. Mount the new O-ring onto the quartz tube (as shown in view **B** in **Figure 10-12** above) and insert the tube into the Unit. Make sure it is centered.

Step 17. Add the second new O-ring on the other side of the quartz tube.

Step 18. Install the quartz tube holder on the side without the UVIS mirror and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**

Figure 10-13: Replacing the Lamp's quartz tube (Generic View 2)

Assembly

Step 19. Install the quartz tube holder on the opposite side – the one with the UVIS mirror.

Step 20. Using a #4 Allen wrench, close the screws in diagonal pattern and tighten them.

Step 21. To reinstall the lamp, perform **Step 11.** to on page **76**.

Powering Up the System

Step 22. Check that the electrical cabinet's power cable is connected to the main power source.

Step 23. Open the Electrical Cabinet and lift all of the circuit breaker switches to the upper **(On)** position.

Step 24. Connect the power cable of the **Controller** to the main power source.

Step 25. On the Main screen of the **Controller**, press

Step 26. Wait 10 minutes for lamps to turn on and the system to stabilize.

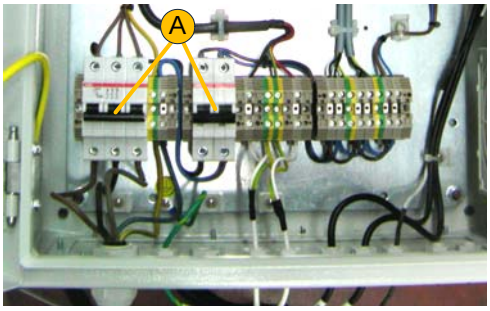


Figure 10-14: Electrical Cabinet Circuit Breakers

Step 27. Check the main screen of the **Controller** and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range (see the table below to enter your expected values).

Notify the executive maintenance engineer in case the values are out of range.

Table 10-1: Abbreviation of Table in Appendix B. Checking the System Parameters

	Parameter	Expected Value	
		Minimum	Maximum
1	Power		
2	Flow Rate		
3	UVT		
4	UV Dose		
5	Lamp Status		

For periodic checks on Atlantium system parameters, a convenient check list is supplied in Appendix **B**, ***Checking the System Parameters*** on page **102**.

10.7 Checking / Replacing Sensors

From time to time, sensors may need to be checked (most frequently for condensation) or replaced. There are two types of sensors. The sensor itself is the same in both types. The difference is the sensor casing and how it is fastened to the Atlantium Unit. The two types are:

- **Adjustable UVIS sensor** - The Sensor's position is adjustable and its casing is fastened with nuts. See below.
- **UVT sensor** - The sensor's casing screws into a fixed position. This type of sensor may contain a pinhole. See ***Replacing/Checking a UVT Analyzer*** on page 85.

10.7.1 Checking an Adjustable UVIS Sensor

Performed as needed

Before removing the UVIS sensor to check it for moisture, do one of the following:

If you do not have a caliper measurement tool: You must mark the placement of the sensor so that you can replace it in the correct position.

If you have a caliper: You must measure the position of the sensor or by using a marker to mark the position before removal.

Prepare the following:

- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

➲ To check an adjustable sensor (using a marker):

Step 1. In the **Controller**, note the value for UVIS for the lamp on which the sensor to be replaced. For details on finding the Controller screen with this information, see the section, ***System Tuning*** on page 86.

Step 2. Mark the placement of the sensor so that you can replace it in the correct position. Use the thin marker to mark the position before removal.

Mark the sensor in two places, a line along the nut and across the nut and perpendicular to the first line. as shown in **(1)**.

Step 3. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

Step 4. When you return the sensor to its place, make sure the thin lines line up to the nut and align together.

Step 5. In the **Controller**, check that the value for UVIS for the lamp on which the sensor was replaced is identical to the value before the sensor was removed. For details on finding the Controller screen with this information, see the section, ***System Tuning*** on page 86.

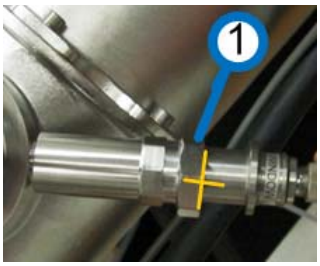


Figure 10-15: Marking the Sensor

➲ To check an adjustable sensor (using a caliper):

Step 1. Place the measurement tool on the UVIS sensor as show in **Figure 10-16** below. Use the screw to adjust the placement of the caliper measurement tool.

Figure 10-16: Measuring the UVIS sensor

Figure 10-17: Caliper Measurement Tool

Step 2. Carefully remove the caliper, taking care not to accidently change its position, and set it aside.

Step 3. Remove the sensor according to the relevant steps in ***Replacing an Adjustable UVIS Sensor*** on page **83**.

Step 4. If you are in the midst of a Calibration process for the UVIS sensor, replace the original UVIS sensor with the Reference sensor.

Step 5. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

Step 6. When you insert a Reference Master sensor or return a sensor to its place, using the set caliper (calibrated to the original UVIS sensor's position) adjust the sensor so that it is set to the position of the original. Refer to **Figure 10-18** to the left.

Figure 10-18: Adjusting the Reference sensor

10.7.2 Replacing an Adjustable UVIS Sensor

Performed as needed

Before removing the UVIS sensor when calibrating it against the Reference Master sensor or checking for moisture, you must mark the placement of the sensor so that you can replace it in the correct position. See above.

Replacing the UVIS sensor involve the loosening of the nut holding it in place.

Prepare the following:

- Replacement UVIS sensor or appropriate type
- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

To replace an adjustable UVIS sensor:

Step 1. To disconnect the harness connector (shown in view **A** in **Figure 10-20**), unscrew the sensor's threaded locking nut.

Step 2. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view **B** below. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 3. Holding the left wrench **#24** steady, twist the right wrench **#30** to loosen the nut.

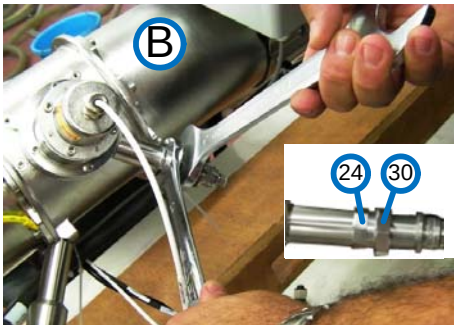


Figure 10-19: Replacing the Sensors (Generic View 2)

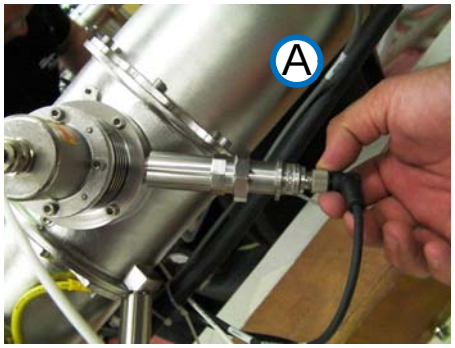


Figure 10-20: Replacing the Sensors (Generic View 1)

Step 4. Carefully pull the sensor out of the housing **(C)**. The Large nut comes off together with the sensor.

Step 5. To enable the system to track relevant data about the sensor, make a note of the serial number of the replacement sensor (for later use).

Step 6. Verify that the sensor lens **(1)** is clean. If needed, using a lens cloth, clean it. (See below.)

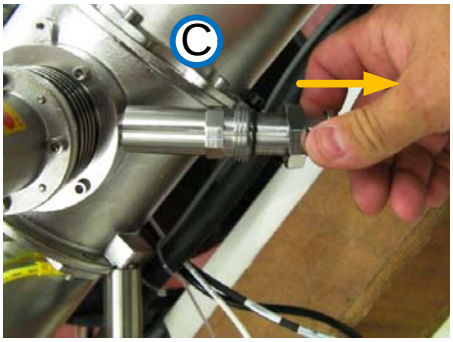


Figure 10-21: Replacing the Sensors (Generic View 3)

Step 7. Position the nut **(2)** on the sensor as shown above. Then position the washer **(3)** on the sensor as shown above.

Step 8. Insert the sensor into the housing.

- If you are replacing the sensor, positioning it about 2/3 the way in. Fasten the nut onto the housing.
- If you are returning a sensor that was removed for checking or calibration, see ***To check a sensor or remove it for calibration*** on page **82**.

Step 9. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view **B** above. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 10. Holding the left wrench **#24** steady, twist the right wrench **#30** to tighten the nut.

Step 11. Reconnect the harness's threaded cap to the new sensor.

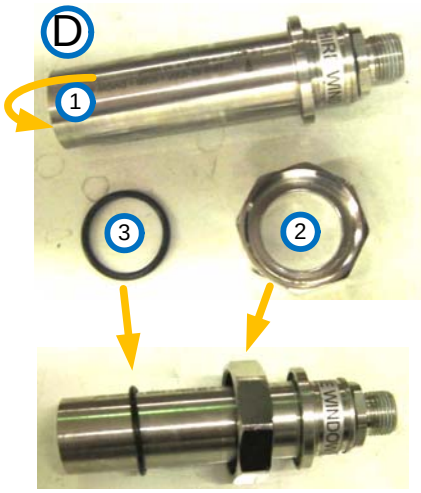


Figure 10-22: Sensor, washer and nut

Step 12. For a replacement sensor, in the **Controller**, In the **Controller**, perform **System Tuning**, see the section, ***System Tuning*** on page **86**.

10.7.3 Replacing / Checking a UVT Analyzer

Performed as needed

For some of the sensors, there is an pinhole located on the sensor's window. This pinhole is calibrated by Atlantium for the specific measurement point. Do not switch disks between ports. Before inserting the sensor, be sure to include the pinhole if present.

Prepare the following:

- Replacement sensor

➲ To replace a sensor:

Step 1. To disconnect the harness connector (shown by the arrow in view **A** below), unscrew the sensor's threaded locking nut.

Step 2. Unscrew the sensor's threaded locking nut from the bottom of the sensor housing, as shown in view **B**.

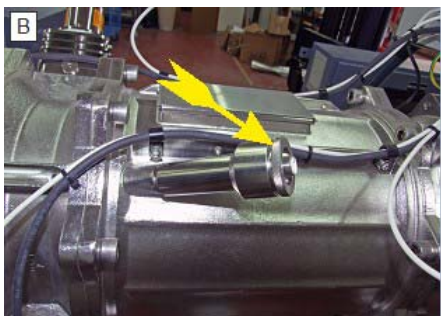


Figure 10-23: Replacing the Sensors 2 (Generic View 2)

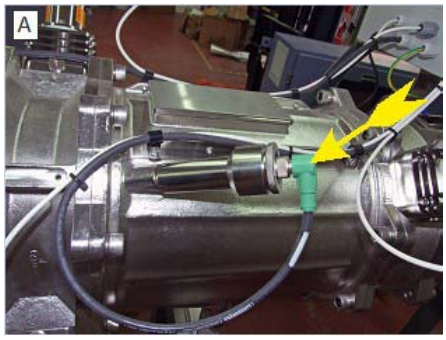


Figure 10-24: Replacing the Sensors 2 (Generic View 1)

Step 3. Carefully pull the sensor out of the housing **(C)**.

Step 4. If an pinhole **(E)** is present it slides out.

Step 5. Unpack the new sensor **(D)** and (if it is supplied) the sensor pinhole.

Figure 10-25: Replacing the Sensors 2 (Generic View 4)

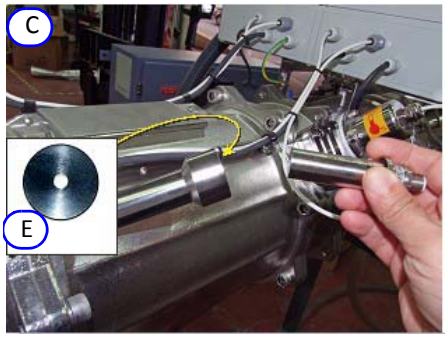


Figure 10-26: Replacing the Sensors 2 (Generic View 3)

Step 6. To enables the system to track relevant data about the sensor, make a note of its serial number.

Step 7. Verify that the sensor lens **(F)** is clean. If needed, using a lens cloth, clean it.

Step 8. If an pinhole **(E)** is present, insert it.

Step 9. Insert the sensor into the housing, and screw the threaded cap back into the housing to secure the sensor.

Step 10. Reconnect the harness's threaded cap to the new sensor.

Step 11. In the **Controller**, perform **System Tuning**, see the section, ***System Tuning*** on page **86**.

10.8 System Tuning

System tuning is performed on these occasions:

- As the last step in system installation
- As the last step when lamps are changed and after 100 operating hours (as needed)
- Upon advice of an Atlantium Service Representative when the system is reset

➲ To tune the Atlantium system:

Step 1. Just prior to tuning, obtain a water sample from the sampling valve on the inlet side of the Atlantium Unit. Test the water sample in an external spectrophotometer. Write down the results. If a spectrophotometer in not available, record the current UVT value shown on the system's display.

Step 2. To shutdown the Atlantium system, on the **Controller** main screen, press ⏻ and wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Step 3. On the Atlantium Unit, locate each UVIS and UVT Intensity Sensor.

Step 4. On each sensor (refer to the figures in the section, ***Checking/Replacing Sensors*** on page **82**):

- **a** Unscrew the locking nut and gently pull out the sensor.
- **b** With a dry and clean lint free cloth, wipe the UVIS sensor's quartz window.
- **c** Reinsert the sensor and replace the locking nut.

Step 5. To turn on the Atlantium system, on the **Controller** main screen, press ⏻ and wait approximately 3 minutes for the **Ready Lamps On** signal to transmit.

Step 6. To access the **Controller's** Settings screen, from the **Main Operations** screen, press . The **Password** screen appears.

Step 7. Using the numbered keys, enter the password (default password is **1357**) and press . The Settings screen appears.

Step 8. To verify that the lamps are already on, for **Lamps**, press ▶. The Lamps Settings screen appears. (Figure **10-5** on page **74**.) Note that the lamps are set to **On**.

Step 9. Press . The settings are saved and you are returned to the Settings screen.

Step 10. For **Mode**, press ▶. The Mode Settings screen appears.

Step 11. For **Power**, using the ▲▼ arrows, turn each lamp up to **100%** power.

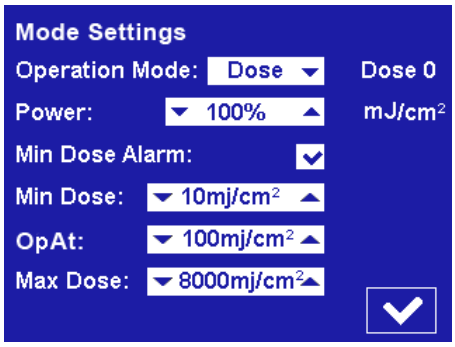


Figure 10-27: Mode Settings Screen

Step 12. Press . The settings are saved and you are returned to the Settings screen.

Step 13. For **Tuning**, press ▶. The System Tuning screen appears.

Step 14. For **UVT**, use the ▲▼ arrows to enter the water test results.

Step 15. For **UVIS 1** and **2** (if your system has two lamps):

- For tuning when replacing a lamp, use the ▲▼ arrows, to enter 100%.
- For tuning with existing lamps, leave the settings as is.

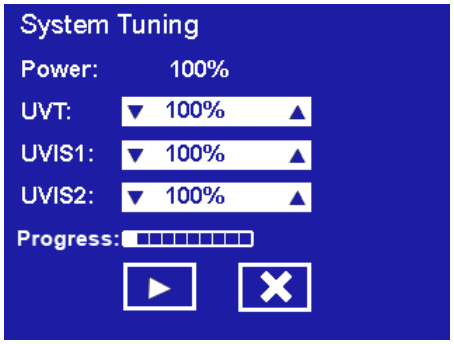


Figure 10-28: System Settings Screen

Step 16. Press . The systems tunes and the **Tune** bar displays the progress.

Step 17. When tuning is complete, the activate arrow turns to . Press it. The settings are saved and you are returned to the Settings screen.

10.9 Cleaning / Replacing the Electrical Cabinet Air Filters

Every three months (on a quarterly basis), the Electrical Cabinet air filter is to be cleaned or replaced as needed. An filter is located on the lower right side of the electrical cabinet The air filter must be replaced once a year regardless of their state.

WARNING!

Where gloves and a mask to protect against breathing dust.

➲ To clean an electric cabinet filter:

Step 1. Using a small, flat screwdriver, on the lower side of the plastic grill **(A)**, press in and disengage the locking tab. Pull the grill off of its frame.

Step 2. Remove the filter from the grill **(B)**.

Step 3. Wash the filter under water with a mild detergent and a soft brush.

Step 4. Let the filter dry. **Make sure it is completely dry before reinstalling it.**

Figure 10-29: Accessing the Air Filter

Step 5. Making sure that the arrows **(C)** on the side of the filter are pointing as shown in **Figure 10-30**, insert the clean filter in the grill and replace the grill into the cabinet's panel.

Step 6. Close the grill and push it in until it snaps into place.

Step 7. Repeat this procedure for each of the additional air filters.

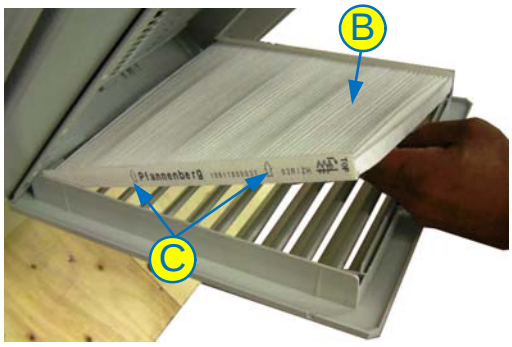


Figure 10-30: Accessing the Air Filter

ATTENTION!

Use the Atlantium replacement filter only - Catalog number **FN0000002**.
Using the wrong filter may cause the system to over heat or allow the entry of water and/or dust.

➲ To replace a filter:

Step 1. Using a small, flat screwdriver, on the lower side of the plastic grill **(A)**, press in and disengage the locking tab. Pull the grill off of its frame.

Step 2. Remove the filter from the grill **(B)**

Step 3. Making sure that the arrows **(C)** on the side of the filter are pointing as shown in **Figure 10-30**, insert the clean filter in the grill and replace the grill into the cabinet's panel.

Step 4. Close the grill and push it in until it snaps into place.

10.10 Advance Maintenance

The advanced maintenance procedures require opening the Electrical Cabinet. **Local regulations may require a certified electrician to perform any of these procedures.**

This section contains:

- ***Replacing the Temperature Sensor*** below.
- ***Replacing the Temperature Converter Card*** on page 90

10.10.1 Replacing the Temperature Sensor

Performed as needed

In the event that the Temperature sensor requires replacement, this section details the procedure.

WARNING!

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Electrical Cabinet or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.
Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the body of the Electrical Cabinet, using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature sensor
- #3 Flat screwdriver
- #2 Phillips screwdriver

To replace the Temperature sensor:

Step 1. Open the Electrical Cabinet and turn the Mains circuit breakers to the **Off** position.

Step 2. Verify with volt meter that there is no power to the Connection card and that no LED is lit on the card.

Step 3. Locate and disconnect the Temperature Sensor connector **(A)**.

Step 4. Detach the wire from the terminal connector and remove the wire from the Electrical Cabinet via the gland at the bottom.

Step 5. Using a #2 Phillips screwdriver, detach the temperature sensor **(C)** from the Atlantium Unit.

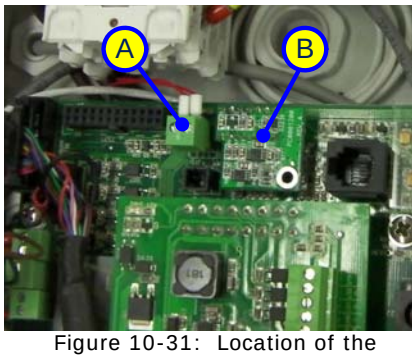


Figure 10-31: Location of the Temperature Sensor Connector

Step 6. Attach the replacement Temperature sensor **(C)** to the Atlantium Unit.

Step 7. Thread the wire through the gland at the bottom of the Electrical Cabinet and up to its terminal connector on the Connection card.

Step 8. Insert the connector **(A)** to its socket on the terminal block.

Step 9. Using plastic fasteners, secure the wire to the anchors and to other cables along the path.

Step 10. In the Electrical Cabinet, turn the Mains circuit breakers to the **On** position.

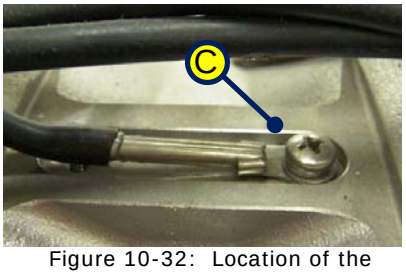


Figure 10-32: Location of the Temperature Sensor on the Atlantium Unit

10.10.2 Replacing the Temperature Converter Card

In the event that the Temperature sensor card requires replacement, this section details the procedure.

WARNING!

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Electrical Cabinet or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.
Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Electrical Cabinet using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature Converter card
- #2 Phillips screwdriver

➲ To replace the Temperature sensor card:

Step 1. Open the Electrical Cabinet and turn the Mains circuit breakers to the **Off** position.

Step 2. Verify with volt meter that there is no power to the Connection card and that no LED is lit on the card.

Step 3. Locate and carefully pull off the Temperature Sensor card **(C)**.

Step 4. Aligning the pins of the replacement Temperature Sensor card **(C)** with the connector (on the board) on the Connection card, carefully insert the small card into the connector of the Connection card.

Step 5. In the Electrical Cabinet, turn the Mains circuit breakers to the **On** position.

11 Troubleshooting

The sections here are comprised of messages headers that appear as error messages in the **Controller.**

Utilize the following safety notes when reviewing the troubleshooting issues:

The solution to the problem might involve exposure to UV light.

The system generates ultraviolet (UV) light within the Atlantium Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working.

Do not look directly into the lamp enclosure during system operation, or while examining or servicing the system's internal components when the system is operating, or when a lamp's caps are open.

The solution to the problem might involve working in a high-voltage environment.

Before replacing a UV lamp or other components, and during any maintenance requiring the lamp to be turned off, **make sure the Electrical Cabinet's circuit breakers are turned OFF.**

Place a sign on the Electrical Cabinet alerting others **NOT** to touch the circuit breakers and touch screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

The solution to the problem might involve working with hot surface.

Do not touch the UV lamp with your bare hands. Lamps can reach a temperature of 1,000°C under operating conditions.

Wait at least 10 minutes until the lamp is cooled down before replacing it.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

Troubleshooting issues

The UV dose is below the allowable threshold on page **11-92**

Unit temperature is too high on page **11-93**

Unit temperature is critically high! on page **11-93**

The rate of the water flow is too low on page **11-94**

UV lamp ignition failed on page **11-95**

Temperature sensor fault on page **11-96**

Ballast temperature is too high on page **11-96**

System type settings error on page **11-97**

Lamp mains failure on page **11-97**

Unexpected failure for an unknown reason on page **11-98**

Lost communication with ballast on page **11-99**

UV lamp ignition failed on page **11-95**

Clean system deposit on page **11-99**

11.1 The UV dose is below the allowable threshold

Controller displays this warning message
UV Dose Below Allowable Threshold

!

	Possible Cause	Solution	
1	The lamp's life has expired.	Check the Lamp's hours displayed on the **Controller's** main screen and replace the lamp if needed. See ***Replacing the Lamp*** on page **10-73.**	
2	The Water flow rate is too high.	Check the water flow on the **Controller's** main screen. If it is higher than normal, decrease the rate of the water flow through the system. See ***Configuring the Flow Settings*** on page **7-53.**	
3	UV lamp efficiency has dropped due to condensation.		Check the UVIS sensors and clean them if necessary.
4	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See ***Cleaning In Place (CIP)*** on page **10-65**. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.	
5	The UVT sensor has malfunctioned.		Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See ***Checking/ Replacing Sensors*** on page **10-82.**
6	The system is operating on **Power mode** instead of **Dose mode**.	If the system settings are in Power mode, in the **Controller**, access the Settings screen and change the **Operation Mode** from **Power** to **Dose** setting. Or, change the Power mode settings to allow higher power to respond to operating conditions. See ***Configuring the Operation Mode*** on page **7-51.**	

11.2 Unit temperature is too high

Controller displays this warning message
Unit Temperature Is Too High

	Possible Cause	Solution
1	The Max temperature value is too low.	Check the system's **Max Temperature** settings. See ***More System Settings*** on page **7-54.**
2	The water flow has significantly decreased.	Check the flow reading in the **Controller**'s main screen to see if it has deviated from normal flow. Check the flow via the flowmeter. Check the water source for a problem.
3	There are air bubbles or the system is not fully filled with water.	Release air via the air valve.
4	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing a designated air release valve at the highest point on the inlet side of the system.

11.3 Unit temperature is critically high!

Controller displays this warning message
Unit Temperature Critically High

	Possible Cause	Solution
1	If this message appeared right after a message indicating loss of communication with the Ballast (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/365°F, then most likely the communication between the **Controller** and the Connection card is lost.	Follow the solutions offered in ***Ballast temperature is too high*** on page **11-96**.
2	The temperature reading dropped to 185°C / 365°F due to a short-circuited in the Temperature sensor, Temperature converter card or Connection Card	Consult Atlantium Technical Support.
3	The Max temperature value is too low.	Check the system's **Max Temperature** settings. See ***More System Settings*** on page **7-54.**
4	There is no water flow.	Check the flow reading in the **Controller**'s main screen to see if it has deviated from normal flow. Check the flow via the flowmeter. Check the water source for a problem.
5	There are air bubbles or the system is not filled with water to capacity.	Release air via the air valve.

	Possible Cause	Solution
6	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing designated air release valve at the highest point on the inlet side of the system.
7	The Temperature jumper is missing.	Verify the existence of the Temperature jumper in its designated location on the Connection card. Open the top cover and locate Jumper **J32** as shown Figure 11-1: Jumper Location marked **A** in the figure below.
8	The Temperature sensor is faulty.	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page **10-89.**
9	The Temperature Converter card is faulty.	If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page **10-90.**

11.4 The rate of the water flow is too low

Controller displays this warning message
Water Flow Rate: Too Low

	Possible Cause	Solution
1	There is an obstruction in the upstream system or pump malfunction.	If this is a planned decrease or temporary stop, change the **Low Flow mode** settings from **Shutdown mode** to **Restart**. See ***Configuring the Flow Settings*** on page **7-53.** This sets the system to Standby. Check pumps, filters and system feed and increase the flow rate. See ***Configuring the Flow Settings*** on page **7-53.**
2	There is no water running through the system.	If this is a planned stop, change the **Low Flow mode** settings from **Shutdown mode** to **Restart**. See ***Configuring the Flow Settings*** on page **7-53.** This sets the system to Standby.
3	The flow meter/flow switch is faulty or a wire has been disconnected.	Check the wires connecting the flow meter/flow switch and reconnect or replace if damaged. Replace the flow meter/flow switch is faulty.

11.5 UV lamp ignition failed

Controller displays this warning message
UV Lamp %d Ignition Failed

	Possible Cause	Solution	
1	The system is set to the wrong model.	In the **Controller**, access **Technician screen #2**. Verify that the system is set to the correct model, number of lamps and branches. See ***Configuring the Technician Settings*** on page **7-55.**	
2	One or more connectors are burnt out.		Turn off the system's main power source Disconnect and thoroughly inspect all of the connectors. IF replacement is needed, contact Atlantium Technical Support.
3	The power supply inside the Electrical Cabinet is faulty.		Replace the Ballast in the Electrical Cabinet with a new one. See ***Cabling Inside the Electrical Cabinet.*** on page **5-41** and Consult Atlantium Technical Support.
4	A lamp is faulty/burned out.		Replace the lamp with a new one. See ***Replacing the Lamp*** on page **10-73.**
5	One or more of the fuses in the Connection card is burned out.		Replace the burned out fuse on the Connection card. Consult Atlantium Technical Support.
6	The Connection card is faulty.		On the Electrical Cabinet and check that the 3-digit display and LEDs are lit. If they are dark, replace the Connection card. Consult Atlantium Technical Support.

11.6 Temperature sensor fault

Controller displays this warning message
Temperature Sensor Fault

	Possible Cause	Solution
1	If this message appeared right after a message indicating loss of communication between components and the temperature display on the monitor is 165°C/365°F, then most likely the communication between the **Controller** and the Connection card is lost.	Follow the solutions offered in ***Ballast temperature is too high*** on page **11-96**.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuited in the Temperature sensor, Temperature converter card or Connection Card	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page **10-89.** If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page **10-90.** If the problem still persists replace the Connection card. Consult Atlantium Technical Support

11.7 Ballast temperature is too high

Controller displays this warning message
Ballast Temp Too high Lamp %d Fault

	Possible Cause	Solution
1	The Connection card environment's temperature is above 80 degrees Celsius.	Check the environment temperature in the Connection card area, open the Electrical Cabinet and look for a possible cause for the heating. See ***Cabling Inside the Electrical Cabinet.*** on page **5-41.**

11.8 System type settings error

Controller displays this warning message
Unset Error Fault at Lamp %d

	Possible Cause	Solution
1	**Lamp Unset:** The System is set to the wrong system model. (Example: System is **RZ163**, but the **Controller** settings read **RZ300**.)	In the **Controller**, access the **Technician screen #2**. See ***Configuring the Technician Settings*** on page **7-55.** Verify that the system is set to the correct system model, number of lamps and branches. If the system is already set to the correct model, you might need to refresh the selection: **Step 1.** Change the model to a different one (e.g. your system is **RS104**, but the selection to **RZ300**). **Step 2.** Exit the **Technician** screen and close the **Controller** circuit breaker. **Step 3.** Open the **Controller** circuit breaker and enter **Technician** screen. **Step 4.** Change the model back to the correct model of your system. **Step 5.** Exit the technician screen and close the **Controller** circuit breaker. **Step 6.** Open the **Controller** circuit breaker and turn the lamps on.

11.9 Lamp mains failure

Controller displays this warning message
Lamp Mains Fail Lamp %d Fault

	Possible Cause	Solution	
1	**Mains Fail issue:** The ballast does not receive enough power from the mains.		A certified electrician is required to verify that the Ballast receives the proper voltage feed from the Mains via the Electrical Cabinet.
2	One or more of the fuses in the Connection card is burned out.		A certified electrician is required to replace the burned out fuse in the Connection card. Consult Atlantium Technical Support
3	A combined problem inside the Ballast regarding power feed and temperature.	Consult Atlantium Technical Support.	

11.10 Unexpected failure for an unknown reason

Controller displays this warning message
Unexpected Failure Unknown Reason %d Fault

	Possible Cause	Solution	
	Lamp Unknown: The lamp was unexpectedly shutdown due to one of the following reasons:		
1	There is a power interruption (for micro seconds) that caused the Ballast to stop working while the Connection Card still held power.	Turn the lamps back on using the On/Off button on the **Controller**'s main screen.	
2	The Lamp burned out during operation.		Replace the Lamp. See ***Replacing the Lamp*** on page **10-73.**
3	The Lamp's connector(s) burned during operation.		Replace the connectors. Consult Atlantium Technical Support.
4	Ballast malfunctioned during operation.		Replace the Ballast. Consult Atlantium Technical Support.

11.11 Lost communication with ballast

Controller displays this warning message
Lost Communication with Ballast

	Possible Cause	Solution
1	**Communication Error:** The communication between the **Controller**'s display card and Connection card was lost while the lamp was on or during the ignition process.	Verify that all Connection card has power and the communication wire is connected properly on both ends. If the problem persist, contact Atlantium support for further instructions.
2	No power is feeding into the Connection card.	Open the Electrical Cabinet and verify that all Connection card has power. See ***Cabling Inside the Electrical Cabinet.*** on page **5-41.**
3	A wire is loose.	Check all cables and tighten connections. See ***Cabling Inside the Electrical Cabinet.*** on page **5-41.**.
4	A jumper is missing.	A communication jumper might need to be installed, contact Atlantium Support for further instruction.
5	Ground is not connected to the **Controller**'s power inlet.	A certified electrician is required to verify that the **Controller** is properly grounded. See ***Cabling Inside the Electrical Cabinet.*** on page **5-41.**
6	There is a Hardware problem in the **Controller**'s display card or in the Connection card.	Contact Atlantium support for further instructions.

11.12 Clean system deposit

Controller displays this warning message
Please Clean System Deposit

		Possible Cause
1	Excessive mineral deposit has collected on the quartz tube.	Perform the Cleaning in Place procedure. See ***Cleaning In Place (CIP)*** on page **10-65.**
2	The **UVT Min** value in the Technician screen is set too high.	See ***Configuring the Technician Settings*** on page **7-55.**

Appendix A Modbus Communication Protocol

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks.

Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

For information on how to configure the communication properties and make the cable connections, see ***Installation Check List*** on page 48.

A.1 Modbus Registry Map

The table on the following pages provides the information you need to help you configure your network to properly communicate with the Atlantium system.

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Registry address	Registry name	Divide	Type	Comment	Units	Read/ Write
100	UVT	:100			%	R
101	Flow	:10			m^3/h	R
102	Dose				mJ/cm^2	R
103	System Power				%	R/W
122	Cleaning System Every				Minutes	R/W
123	Cleaning System Duration				Minutes	R/W
125	Op At dose	:10			mJ/cm^2	R/W
126	Minimum Dose	:10			mJ/cm^2	R/W
128	Maximum Dose	:10			mJ/cm^2	R/W
129	Maximum Flow			Flow value while in "Flow Switch" mode		R
0	System On/Off		Coil Bit	0 -> Off, 1 -> On		R/W
1	Cleaning System Enable/ Disable		Coil Bit	0 -> Enable, 1 -> Disable		R
2	Operation Mode Dose/ Power		Coil Bit	0 -> Dose, 1 -> Power		R
4	Remote On/Off Enable/ Disabled		Coil Bit	0 -> Enable, 1 -> Disable		R
6	Alarm/Ready output		Coil Bit	0 -> Alarm, 1 -> Ready		R
7	Flow Switch/Flow Meter		Coil Bit	1 -> Switch, 0 -> Meter		R
8	Flow Switch Polarity		Coil Bit	0 -> NO, 1 -> NC		R
9	Dose < min dose		Coil Bit	1 -> OK, 0 -> Fault		R
11	Over Temp		Coil Bit	1 -> OK, 0 -> Fault		R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Registry address	Registry name	Divide	Type	Comment	Units	Read / Write
14	Min flow alarm		Coil Bit	1 -> OK, 0 -> Fault		R
16	General alarm		Coil Bit	1 -> OK		R
17	Cleaning System exist		Coil Bit	0 -> Exist		R
18	Critical Over Temp		Coil Bit	1 -> OK		R
19	System Ready		Coil Bit	1 -> Raedy		R
21	System status		Coil Bit	(Off/On)		R
22	Number of lamps		Coil Bit	0 -> 1 lamp, 1 -> 2 lamps		R/W
23	Lamp 1 Enable/Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
24	Lamp 2 Enable/Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
Lamp 1						
106	Status lamp 1				%	R
107	UVT lamp 1				%	R
108	Age hours lamp 1 MSB				Hours	R
109	Age hours lamp 1 LSB				Minutes	R
110	Unit Temp	:100			°C	R
111	Power lamp 1	:1000			KW	R
112	UVIS lamp 1	:100			mA	R
113	UVTS lamp 1	:100			mA	R
132	LM Card 1 Temp				°C	R
Lamp 2						
114	Status lamp 2				%	R
115	UVT lamp 2				%	R
116	Age hours lamp 2 MSB				Hours	R
117	Age hours lamp 2 LSB				Minutes	R
119	Power lamp 2	:1000			KW	R
120	UVIS lamp 2	:100			mA	R
121	UVTS lamp 2	:100			mA	R
133	LM Card 2 Temp				°C	R

Appendix B Checking the System Parameters

You can use the checklist below to keep a record of your checks on Atlantium system parameters. **Print this page**.

Table B-1: Atlantium System Parameters Checklist

Date	Time	Lamp ID	Power	Flow Rate	UVT	UV Dose	Lamp Status	OK Y/N	Remarks/Alarms/ Events
			Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____		

Appendix C Glossary of Terms

Term	Definition
Auxiliaries	Valves, flow devices, etc. essential to the workings of the Atlantium system but not directly a part of the Atlantium system.
Ballast	The power supply that translates the electrical power to UV power.
Connection Box	The box that sits on top of each lamp unit and holds the circuit board for the Electrical Cabinet or Ballast Module electrical configurations. Not relevant for Ballast Module/Cable-style.
Controller	Controller that controls operations and dose measurement. Comes in two versions: Standard and Premium. The Standard has a monochromatic screen and several key features. The Premium has a color screen can accommodate a full complement of reporting and trending features.
Cooling System	The system that controls the cooling of the lamp to assure its optimal operation.
Quartz Chamber	The **quartz pipe** in which the disinfection occurs.
External On/Off	This enables an external source (SCADA, PLC, Process) to turn the Atlantium system on and off.
Atlantium provides a choice of electrical configurations. Electrical configuration styles:	
Electrical Cabinet-style (EC)	**EC** - Electrical Cabinet.
Ballast Module (C)	**C -** Waterproof case for the ballast/circuit board combination with direct wire connections to each component.
Ballast Module/Plug-style (P)	**P** -: Waterproof case for the ballast which connects via three plugs from the circuit board which sits on top of each lamp unit.
Inlet/Outlet Valves	Isolating Valves on the Inlet/outlet sides.
Lamp Chamber	The chamber which holds the lamp.
Lifting Support	The lifting element which can also be used to help support the Unit.
Pinhole	This is a small disc that is fitted for each sensor port to assure that the right amount of light penetrates.
Portable CIP System	The portable unit Atlantium sells that connects to the Unit to clean and disinfect it without disassembling. It includes several kits – a valve kit, an accessories kit and the CIP kit itself.
General Alarm	This enables the Atlantium unit to send an alarm to a buzzer, cell phone or other destination.

Term	Definition
Triangle Support Brackets	RZ-Series only- brackets that can be used to support and balance a unit.
System Ready Signal	The signal the Atlantium unit provides to tell the production system that the machine is ready to supply properly disinfected water.
System Tuning	This is process is performed when key components are replaced or changed to synchronize them.
TRAC Software	**T**otal **R**eporting **A**nalysis & **C**ontrol Software System; the software application that controls dose measurement and maintenance, and stores, analyzes and reports real time; enables viewing and controlling operations remotely for up to 9 Atlantium systems on one TRAC System. Provides full validated control and One-Touch pre-programmed reports for regulatory compliance with EPA and PMO rules.
Ultrasonic Cleaner	Ultrasonic cleaning system.
UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds the real time dose calculation.
UVIS Calibration	This is the EPA required process to check that the UV Intensity Sensor is reading correctly and has not drifted beyond the 20% tolerance allowed by the UVDGM.
UVT Sensor	The sensor that measures the lamp UV intensity through the water. Together with the UV Intensity sensor and the Controller, it analyzes, tracks and reports the water quality.
UVT Sensor Calibration	This is the EPA required process to check that the UVT Sensor is reading within 2% of the actual UVT.
Validation Envelope	This is the matrix of flow, power and UVT values that are covered by the EPA validation.